

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_3d035317-127\agent-ace7aafd0ff15b481.jsonl`
- SHA-256: `9f937fc19704013feab906e7a6be28a9f94a7c2fd82ae622d5664dfbfc843e92`
- Source modified: `2026-06-22T09:48:48+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_3d035317-127`
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-22T09:46:23.309Z)

You are fixing staleness in ONE doc of the KhelSutra repo, applying an audit that already ran. Be SURGICAL: minimal, targeted edits that match the audited fix; preserve the doc's voice, structure, and markdown formatting. Do NOT wholesale-rewrite a doc.

KhelSutra repo root: C:\Users\avidu\Projects\kheLSutra
ENGINE repo root (sibling, for cross-repo verification): C:\Users\avidu\Projects\badminton-highlight-indexer (origin/master freshly fetched)
Today: 2026-06-22. KhelSutra is on branch docs/freshness-sweep (off master).

YOUR DOC (edit ONLY this file):
C:\Users\avidu\Projects\kheLSutra\docs\MULTILINGUAL_EXPANSION.md

The audit findings are in this JSON file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\tasks\w3dbzrpby.output
Shape: { result: { perDoc: [{doc, overallStatus, findings, summary}], staleFindings: [{doc, claim, docLocation, verdict, evidence, suggestedFix, severity}] } }.
Read it and select ONLY entries where doc === "docs/MULTILINGUAL_EXPANSION.md". Your worklist = that doc's staleFindings. Also skim that doc's perDoc.findings for "unverifiable" items ? do NOT act on those unless you can now positively verify them.

For each stale finding:

1. RE-VERIFY it yourself against the LIVE repo before editing ? Read the doc lines + the cited code. For any claim about the ENGINE (paths like backend/main.py, /api/*, native_wasb_runner.py, auth/CORS), verify against the engine repo's origin/master: e.g. ``git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "<symbol>" origin/master -- backend/`` or ``git -C ... show origin/master:backend/main.py``. Trust nothing blindly ? the audit itself may be slightly off and engine line numbers DRIFT between PRs.
2. If confirmed, apply the suggestedFix as a SURGICAL Edit. For brittle ENGINE line-number anchors, prefer SYMBOL-based references (e.g. ``backend/main.py::get_config``) over chasing exact line numbers ? but only after confirming the symbol exists on engine origin/master.
3. DOCS ONLY: if a fix would require a CODE change (e.g. a comment in site/server.js), do NOT edit code ? record it under codeChangesNeeded.

Hard rules:

- Edit ONLY your assigned doc file. Run NO git WRITE commands (no add/commit/checkout/branch/stash) ? read-only git for verification only.
- Keep status markers HONEST: flip Todo->Done only WITH the PR number / file evidence; never invent a completion.
- If a finding is wrong, not reproducible, or too risky to auto-apply, SKIP it and explain under skipped.
- Do not "improve" beyond the findings; no scope creep.

Return the structured report.

2. assistant (2026-06-22T09:46:25.701Z)

I'll start by reading the audit findings and my assigned doc.

3. user (2026-06-22T09:46:28.068Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-591 of 1722 total (61859 tokens, cap 25000). Call Read with offset=592 limit=591 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      {
2      "summary": "Audit every KhelSutra doc for staleness against the live repo, returning
structured findings",
3      "agentCount": 15,
4      "logs": [
5      "audited 15/15 docs; 44 stale findings; archive-candidates: none"
6      ],
7      "result": {
8      "perDoc": [
9      {
10     "doc": "NEXT_STEPS.md",
11     "overallStatus": "minor-staleness",
12     "summary": "NEXT_STEPS.md is overwhelmingly accurate against master: routes,
file paths, the Worker/D1 names, the 6-language localization, the 3-step wizard, the per-rally
PR numbers (#11/#14/#21/#28/#30/#31), the appliance/per-rally doc statuses, and every still-
pending \"Queued\" item all check out against code and git. The only real staleness is the \"PRs
through #93 merged\" count, which lags by one because PR #94 (the very commit that wrote
\"through #93\") merged afterward; plus a low-severity GCS-artifact filename mismatch. Not an
archive candidate ? it is the live next-steps tracker.",
13     "findings": [
14     {
15     "claim": "PRs through #93 merged (master).",
16     "docLocation": "NEXT_STEPS.md:13",
17     "verdict": "stale",
18     "evidence": "git log --oneline -1 = '50b2bd7 chore(repo): ... correct stale
NEXT_STEPS claims (#94)'; total 94 commits/PRs on master. Ironically #94 is the commit that
updated this line to '#93', then merged itself, so the doc now lags one PR behind master.",
19     "suggestedFix": "Update to 'PRs through #94 merged (master).'",
20     "severity": "medium"
21     },
22     {
23     "claim": "Where we are (2026-06-20): ... khelsutra.guru is LIVE and fully
shipped (landing + /capture checklist + OG image + segmented Request-Access form).",
24     "docLocation": "NEXT_STEPS.md:5-14",
25     "verdict": "correct",
26     "evidence": "site/server.js:98 maps /capture->capture.html;
site/public/og.png + capture.html tracked; index.html:191-193 textareas + lines 537-581
academy/individual/gpu/storage fields. Header date 2026-06-20 is slightly behind the latest work
(PR #94 merged 2026-06-22) but the substance holds.",
27     "suggestedFix": "Optionally bump the 'Where we are' date to 2026-06-22 to
match the latest merged work.",
28     "severity": "low"
29     },
30     {
31     "claim": "On Cloudflare (Worker hidden-king-khelsutra + D1 khelsutra-
access): the form persists to D1 and emails a summary via an Email Routing send binding.",
32     "docLocation": "NEXT_STEPS.md:7-9",
33     "verdict": "correct",
34     "evidence": "site/wrangler.toml: name = \"hidden-king-khelsutra\";
[[d1_databases]] database_name = \"khelsutra-access\"; [[send_email]] name = \"NOTIFY\".
site/src/index.js:80 handles POST /api/request-access -> insert into D1 binding DB.",
35     "suggestedFix": "",
36     "severity": "low"
37     },
38     ]
39     },
40     }
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38         {
39             "claim": "CI gates every PR ? three jobs: full site npm test, web/ SPA build
+ coverage floor, and a real-browser layout check.",
40             "docLocation": "NEXT_STEPS.md:9-12",
41             "verdict": "correct",
42             "evidence": "site/package.json scripts include test (node --test),
test:layout (layout.browser.mjs); web has a coverage floor (#80). PR #94 message confirms 'three
jobs: site npm test = 86 green, web SPA build+coverage, real-browser layout'. Non-browser site
suite ran green (61 tests in the 4 non-layout files).",
43             "suggestedFix": "",
44             "severity": "low"
45         },
46         {
47             "claim": "The form also runs on a dependency-free Node + SQLite host
(site/server.js), live on the DO droplet -> http://168.144.126.176:8787 (systemd khelsutra-
site).",
48             "docLocation": "NEXT_STEPS.md:11-13",
49             "verdict": "correct",
50             "evidence": "site/server.js exists, uses node:sqlite (header comment line
2). IP 168.144.126.176 + port 8787 + 'khelsutra-site' all appear in
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48 ('/srv/khelsutra-site, node :8787') and deploy refs.
Live-status is external/unverifiable from the repo, but the host code and identifiers are
correct.",
51             "suggestedFix": "",
52             "severity": "low"
53         },
54         {
55             "claim": "Docs: docs/REQUEST_ACCESS_FORM.md, docs/PORTABILITY.md.",
56             "docLocation": "NEXT_STEPS.md:13-14",
57             "verdict": "correct",
58             "evidence": "git ls-files: both docs/REQUEST_ACCESS_FORM.md and
docs/PORTABILITY.md are tracked.",
59             "suggestedFix": "",
60             "severity": "low"
61         },
62         {
63             "claim": "DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22): localized to
all 6 languages (en/kn/hi/es/da/id); full record docs/archives/FRIENDLY_AND_MULTILINGUAL.md.",
64             "docLocation": "NEXT_STEPS.md:17-21",
65             "verdict": "correct",
66             "evidence": "docs/archives/FRIENDLY_AND_MULTILINGUAL.md is tracked; git log
shows PR #93 'docs: archive the Friendly & Multilingual tracker'. web/src/i18n/index.ts:25
SUPPORTED_LOCALES = [en,kn,hi,es,da,id]; PR #87 added es/da/id to the marketing site.",
67             "suggestedFix": "",
68             "severity": "low"
69         },
70         {
71             "claim": "Interactive per-rally selection MVP/M0 COMPLETE; Done: P0 design
(#11), P1 scaffold (#14), P2 grid+hook (#21), P3 parseQuery query bar (#28), P4 SelectionFooter
(#30), P5 real load+download (#31). Tracked doc docs/PER_RALLY_SELECTION.md (IN PROGRESS).",
72             "docLocation": "NEXT_STEPS.md:32-41",
73             "verdict": "correct",
74             "evidence": "git log: cb74ce5 #11 'tracked design ... RallyPicker'; aac7e0c
#14 'P1 scaffold'; 62598e7 #21 'P2 selectable rally grid + hook'; d537d19 #28 'P3 deterministic
parseQuery() query bar'; aff8de8 #30 'P4 SelectionFooter + min_shot_count warning'; 8bf200d #31
'P5 load real rallies + Build->compile->preview->download'. docs/PER_RALLY_SELECTION.md status
header = 'IN PROGRESS ? MVP/M0 (P1-P5) COMPLETE'.",
75             "suggestedFix": "",
76             "severity": "low"
77         },
78         {
79             "claim": "P2c first-run setup wizard ? 3-step SetupWizard from
/api/capabilities, 6 languages.",
80             "docLocation": "NEXT_STEPS.md (implied via track) and verified for the
merged #88",

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81         "verdict": "correct",
82         "evidence": "web/src/components/SetupWizard.tsx:22 'const STEPS = 3'; line 1
comment 'Three quick steps'; line 3 'Localized (en/kn/hi/es/da/id)'; web/src/api/client.ts:139
GET /capabilities. git: 7ae9cee #88.",
83         "suggestedFix": "",
84         "severity": "low"
85     },
86     {
87         "claim": "Single-box appliance ? Tracked design
docs/LOCAL_APPLIANCE_ARCHITECTURE.md (DRAFT, owner-gated).",
88         "docLocation": "NEXT_STEPS.md:52-58",
89         "verdict": "correct",
90         "evidence": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md status header: 'Status:
DRAFT ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5 bridge)'.",
91         "suggestedFix": "",
92         "severity": "low"
93     },
94     {
95         "claim": "Queued: Always-Use-HTTPS + HSTS (Cloudflare) and Cloudflare Web
Analytics token paste into site/public/index.html ? both still pending.",
96         "docLocation": "NEXT_STEPS.md:44-45",
97         "verdict": "correct",
98         "evidence": "No Strict-Transport-Security/HSTS header found in site/*.js or
wrangler.toml. site/public/index.html:629 still has data-cf-beacon token 'REPLACE_WITH_CF_TOKEN'
(commented-out block). Both correctly listed as not-yet-done.",
99         "suggestedFix": "",
100        "severity": "low"
101    },
102    {
103        "claim": "Droplet -> production: add VM-path SMTP so the Node host also
emails (currently noopNotify).",
104        "docLocation": "NEXT_STEPS.md:46-47",
105        "verdict": "correct",
106        "evidence": "site/server.js:14 imports noopNotify; line 24 'const notify =
noopNotify(); // VM host: no email yet (add SMTP later)'.",
107        "suggestedFix": "",
108        "severity": "low"
109    },
110    {
111        "claim": "Org consolidation DONE; khelsutra (this site repo) transfer is an
owner step still PENDING (unchecked).",
112        "docLocation": "NEXT_STEPS.md:65-73",
113        "verdict": "correct",
114        "evidence": "git remote -v shows origin =
https://github.com/avidullu/khelsutra.git (NOT under Khelsutra org) ? consistent with the '?
owner step pending' marker. Other repos already moved: docs/DRY_RUN_CUJ.md:49 links
github.com/Khelsutra/badminton-highlight-indexer#289.",
115        "suggestedFix": "",
116        "severity": "low"
117    },
118    {
119        "claim": "A real-UI CUJ screencast exists -> promo_cuj.mp4/.gif in
gs://khelsutra-rally-corpus/promo/.",
120        "docLocation": "NEXT_STEPS.md:29-31",
121        "verdict": "unverifiable",
122        "evidence": "No GCS access to confirm the object. The companion doc
docs/DRY_RUN_CUJ.md:85 names the actual artifacts 'khelsutra-bucket-cuj-screencast.mp4' and ':60
khelsutra-i18n-screencast.mp4' in the same gs://khelsutra-rally-corpus/promo/ prefix;
'promo_cuj.mp4' appears nowhere else in the repo, suggesting a possible filename mismatch.",
123        "suggestedFix": "Reconcile the filename with docs/DRY_RUN_CUJ.md (likely
'khelsutra-bucket-cuj-screencast.mp4') or confirm 'promo_cuj.mp4' is a real object in the
bucket.",
124        "severity": "low"
125    },
126    {

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127         "claim": "Control plane (control-plane/) at beta.",
128         "docLocation": "NEXT_STEPS.md:62",
129         "verdict": "correct",
130         "evidence": "control-plane/ directory does not exist in the repo (test -d
looking/aspirational item, not a claim of present existence ? so it is not stale as written.",
131         "suggestedFix": "",
132         "severity": "low"
133     }
134 ]
135 },
136 {
137     "doc": "README.md",
138     "overallStatus": "significant-staleness",
139     "summary": "The repo-structure list, deploy references, and canonical-doc
pointers are all correct, but the README's headline status is badly out of date: it still calls
the repo a \"pre-alpha scaffold\" whose SPA has yet to land \"per milestone M1\", whereas the
web/ SPA is in fact built, multilingual, and shipped (per-rally MVP P0?P5, P2c wizard, in-UI
upload, 6 languages; khelsutra.guru is LIVE). The control-plane/ directory it lists also does
not yet exist.",
140     "findings": [
141     {
142         "claim": "Status: pre-alpha scaffold. SPA lands in `web/` per execution-plan
milestone M1.",
143         "docLocation": "README.md:9",
144         "verdict": "stale",
145         "evidence": "web/src/ has 51 tracked source files incl. web/src/App.tsx;
`git log --oneline -- web/` = 31 commits; merged PRs show the SPA is feature-complete: P2c
first-run wizard (#88), in-UI file upload T4 (#66), per-rally selection MVP P0?P5
(#11/#14/#21/#28/#30/#31), 6-language catalogs (#67). NEXT_STEPS.md:7 states 'khelsutra.guru is
LIVE and fully shipped'. UI_ALPHA_EXECUTION_PLAN.md:36 defines M1 only as the initial
'Vite+React+TS+Tailwind scaffold', which is long past.",
146         "suggestedFix": "Replace with a current status, e.g.: 'Status: alpha ? the
web/ SPA is built and multilingual (6 languages), with the per-rally selection MVP and first-run
setup wizard shipped; khelsutra.guru / app.khelsutra.guru are live (Cloudflare Pages +
cloudflared tunnel + Access). See NEXT_STEPS.md.'",
147         "severity": "high"
148     },
149     {
150         "claim": "the web app (alpha SPA) now, the hosted control plane at beta",
151         "docLocation": "README.md:4",
152         "verdict": "stale",
153         "evidence": "This tagline pairs with line 9's 'pre-alpha scaffold', which is
contradicted by the shipped, live SPA (see web/ evidence above and NEXT_STEPS.md:7). The 'alpha
SPA now' framing is directionally right but undercut by the stale 'pre-alpha scaffold' status it
sits beside.",
154         "suggestedFix": "Keep 'web app (alpha SPA) now' but align the rest of the
intro/status so it no longer describes the SPA as not-yet-landed.",
155         "severity": "medium"
156     },
157     {
158         "claim": "`control-plane/` ? (beta) Cloud Run FastAPI control plane",
159         "docLocation": "README.md:17",
160         "verdict": "stale",
161         "evidence": "No control-plane/ directory exists: `ls control-plane` -> 'No
such file or directory'; `git ls-files control-plane/` returns nothing. Top-level tracked dirs
are deploy/, docs/, locales/, site/, web/ only.",
162         "suggestedFix": "Mark it as a future/planned directory that does not yet
exist (e.g. move it out of the 'Structure' list into a 'Planned (beta)' note), so a reader
doesn't expect the path to be present.",
163         "severity": "medium"
164     },
165     {
166         "claim": "Engine ... sibling repo `badminton-highlight-indexer` (rename
pending).",

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167         "docLocation": "README.md:6",
168         "verdict": "correct",
169         "evidence": "Sibling repo still exists under that name: `ls
C:/Users/avidu/Projects/` shows 'badminton-highlight-indexer'; no rename has occurred, so
'rename pending' holds.",
170         "suggestedFix": "",
171         "severity": "low"
172     },
173     {
174         "claim": "Canonical planning docs live in the engine repo's docs/ for now:
UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md.",
175         "docLocation": "README.md:7-8",
176         "verdict": "correct",
177         "evidence": "All three exist in the engine repo (badminton-highlight-
indexer/docs/UI_PROPOSAL.md, HOSTING_PLAN.md, UI_ALPHA_EXECUTION_PLAN.md present) and are NOT
tracked in khelsutra/docs (`git ls-files docs/` in khelsutra returns none of them; `ls
docs/UI_PROPOSAL.md` -> not found).",
178         "suggestedFix": "",
179         "severity": "low"
180     },
181     {
182         "claim": "`deploy/` ... alpha deploy kit ... DEPLOY_ALPHA.md ... +
cloudflared.config.example.yml",
183         "docLocation": "README.md:15",
184         "verdict": "correct",
185         "evidence": "git ls-files deploy/ shows deploy/DEPLOY_ALPHA.md and
deploy/cloudflared.config.example.yml (plus deploy/droplet/*). DEPLOY_ALPHA.md:1-5 matches the
described 'shareable page, no SSH; SPA on Pages + engine on api.* tunnel + CF Access' content.",
186         "suggestedFix": "",
187         "severity": "low"
188     },
189     {
190         "claim": "`site/` ? public marketing landing page (static; served at
khelsutra.guru)",
191         "docLocation": "README.md:13",
192         "verdict": "correct",
193         "evidence": "site/ exists with site/public/index.html, server.js, etc.;
DEPLOY_ALPHA.md and NEXT_STEPS.md confirm the marketing site is served at khelsutra.guru (apex)
while the SPA is at app.khelsutra.guru.",
194         "suggestedFix": "",
195         "severity": "low"
196     },
197     {
198         "claim": "`web/` ? React SPA (alpha; Vite + React + TS + Tailwind;
Cloudflare Pages build root)",
199         "docLocation": "README.md:14",
200         "verdict": "correct",
201         "evidence": "web/ is a Vite+React+TS+Tailwind SPA (web/README.md:3 'Vite +
React + TS + Tailwind v4'); DEPLOY_ALPHA.md:14 confirms 'app.khelsutra.guru -> Cloudflare Pages
(the web/ SPA)'. The tech-stack/build-root facts are accurate; only the inline 'alpha' maturity
ties into the stale status above.",
202         "suggestedFix": "",
203         "severity": "low"
204     },
205     {
206         "claim": "License: All rights reserved (private; no OSS license).",
207         "docLocation": "README.md:19-21",
208         "verdict": "correct",
209         "evidence": "No LICENSE file is tracked (not present in `git ls-files`);
consistent with a private 'all rights reserved' repo.",
210         "suggestedFix": "",
211         "severity": "low"
212     }
213 ]
214 },

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215     {
216         "doc": "deploy/DEPLOY_ALPHA.md",
217         "overallStatus": "minor-staleness",
218         "summary": "Verified correct except a stale header",
219         "findings": [
220             {
221                 "claim": "header stale deploy live",
222                 "docLocation": "deploy/DEPLOY_ALPHA.md:3",
223                 "verdict": "stale",
224                 "evidence": "droplet/README.md:100 live-verified 2026-06-22",
225                 "suggestedFix": "set header to LIVE",
226                 "severity": "medium"
227             },
228             {
229                 "claim": "host list too narrow",
230                 "docLocation": "deploy/DEPLOY_ALPHA.md:66",
231                 "verdict": "stale",
232                 "evidence": "engine.env.example adds loopback",
233                 "suggestedFix": "add localhost",
234                 "severity": "low"
235             }
236         ]
237     },
238     {
239         "doc": "deploy/droplet/README.md",
240         "overallStatus": "fresh",
241         "summary": "This command-dense droplet quickstart is accurate and matches the
live repo: every cited kit file exists with the described contents, and the engine-behavior
claims (fail-fast boot posture, RALLY_* exposure vars, the Gemini-default-on-CPU gotcha, the
gs:// jail rejection, the --server entryptpoint) all verify against the engine repo and
DEPLOY_ALPHA.md. The only items not locally reproducible are the 2026-06-22 droplet live-run
figures (e.g. \"1313 passed\"), which are an off-box assertion ? they are internally plausible
(engine has ~1269 test functions before parametrization) but cannot be confirmed from these
repos.",
242         "findings": [
243             {
244                 "claim": "Four kit files exist: config.json.example, engine.env.example,
khelsutra-engine.service (and ../cloudflared.config.example.yml), each going to the documented
destination with the documented purpose.",
245                 "docLocation": "deploy/droplet/README.md:9-14",
246                 "verdict": "correct",
247                 "evidence": "git ls-files deploy/ lists all four:
deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service} and
deploy/cloudflared.config.example.yml; contents read and match (config.json.example has the
security block + cf_team/aud; engine.env.example has the two RALLY_* exposure vars +
RALLY_APP_HOME; the service binds 127.0.0.1:8000; cloudflared.config routes api. ->
127.0.0.1:8000).",
248                 "suggestedFix": "",
249                 "severity": "low"
250             },
251             {
252                 "claim": "The boot fails fast (exit 1, '[STARTUP] refusing to start') if
edge_auth_enabled=true but the jail or cf_team_domain/cf_access_aud is unset.",
253                 "docLocation": "deploy/droplet/README.md:53-54",
254                 "verdict": "correct",
255                 "evidence": "backend/main.py:857 logs '[STARTUP] refusing to start';
backend/config/startup.py:109 EXPOSED_NO_VIDEO_ROOT and :123 EXPOSED_EDGE_AUTH_INCOMPLETE are
both LEVEL_ERROR (startup.py:41,272-276), while EXPOSED_AUTH_EXTRA_MISSING (PyJWT) is LEVEL_WARN
(:135). From engine PR #294 (commit 0c0e064).",
256                 "suggestedFix": "",
257                 "severity": "low"
258             },
259             {
260                 "claim": "indexing.default_segmenter MUST be gemini on a CPU box; the
shipped default yolo_hybrid needs torch/GPU and every job fails 'No module named torch' without

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it.",
261         "docLocation": "deploy/droplet/README.md:109-113",
262         "verdict": "correct",
263         "evidence": "config.json.example:9-11 sets default_segmenter=gemini. Engine
AppConfig default is yolo_hybrid (backend/config/models.py:128), so an all-defaults load on the
droplet (which hand-writes config.json rather than running the --app LIGHT seeder) does land on
yolo_hybrid. The LIGHT 'motion' default (loader.py:227) only applies via
write_light_default_config (indexer --app first-run, PR #310), not this systemd path ? so the
gotcha is accurate for this deploy.",
264         "suggestedFix": "",
265         "severity": "low"
266     },
267     {
268         "claim": "RALLY_ALLOWED_HOSTS is the load-bearing fix; with edge_auth on,
the Host-header guard defaults loopback-only and 400s every tunneled request until set.",
269         "docLocation": "deploy/droplet/README.md:92, engine.env.example:14-17",
270         "verdict": "correct",
271         "evidence": "backend/config/startup.py:143-147 raises
EXPOSED_HOST_ALLOWLIST_LOOPBACK (LEVEL_WARN) when edge_auth is ON and RALLY_ALLOWED_HOSTS is
unset, with the exact 'will 400 forwarded requests' rationale; backend/main.py reads
RALLY_ALLOWED_HOSTS / RALLY_BIND_HOST / RALLY_CORS_ALLOW_ORIGINS. Matches DEPLOY_ALPHA.md §2.2
repro (400->200).",
272         "suggestedFix": "",
273         "severity": "low"
274     },
275     {
276         "claim": "Pasting a gs:// URI is rejected by the jail on the exposed box;
upload is the alpha path.",
277         "docLocation": "deploy/droplet/README.md:96-98",
278         "verdict": "correct",
279         "evidence": "tests/test_api_endpoints.py:608
test_process_hosted_mode_rejects_nonlocal_uri pins the gs:// -> 400 behavior under
allowed_video_root (referenced at :541 as the counterweight); the jail is required to expose
(config.json.example sets allowed_video_root).",
280         "suggestedFix": "",
281         "severity": "low"
282     },
283     {
284         "claim": "Pages build: Root dir web, Build npm run build, Output web/dist,
with VITE_API_BASE = https://api.khelsutra.guru/api.",
285         "docLocation": "deploy/droplet/README.md:73-76",
286         "verdict": "correct",
287         "evidence": "khelsutra web/ SPA exists (git ls-files web/...);
web/package.json build script = 'vite build'; web/src/api/client.ts:25 reads
import.meta.env.VITE_API_BASE (default '/api'), and client.test.ts:32 exercises the exact split
value https://api.khelsutra.guru/api. Vite default outDir 'dist' relative to root web/ =
web/dist.",
288         "suggestedFix": "",
289         "severity": "low"
290     },
291     {
292         "claim": "The shipped engine is run via the auth extra and python -m
backend.main --server, binding 127.0.0.1:8000.",
293         "docLocation": "deploy/droplet/README.md:33,47-49; khelsutra-
engine.service:26",
294         "verdict": "correct",
295         "evidence": "backend/main.py:867 defines the --server flag and :1107 calls
uvicorn.run(app, host=bind_host, port=port); --setup flag at :865; the service ExecStart uses
python -m backend.main --server.",
296         "suggestedFix": "",
297         "severity": "low"
298     },
299     {
300         "claim": "Live-verified on the droplet (2026-06-22): full suite 1313 passed
+ ruff/mypy clean on the box; boot-posture, Host-guard 400->200, path-jail, T4 upload->jail,

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edge-auth 401-gate exercised live; full DNS->Cloudflare->Access->tunnel->engine chain
confirmed.",
301         "docLocation": "deploy/droplet/README.md:100-106",
302         "verdict": "unverifiable",
303         "evidence": "This is an off-box live-run assertion that cannot be reproduced
from the khelsutra or engine repos here. Sanity check: the engine tests/ dir has ~1269 'def
test_' functions (parametrize expansions push the collected/passed count higher), so '1313
passed' is plausible and consistent, not contradicted. Marked unverifiable per the file:line-
evidence discipline; no sign of staleness.",
304         "suggestedFix": "",
305         "severity": "low"
306     },
307     {
308         "claim": "Cost: Cloudflare Pages + Tunnel + Access are free (Access <= 50
users); standing cost ~ $0 + Gemini per run.",
309         "docLocation": "deploy/droplet/README.md:16-18",
310         "verdict": "unverifiable",
311         "evidence": "External Cloudflare pricing claim; not checkable against the
repo. Consistent with DEPLOY_ALPHA.md:3 ('~$0/mo standing ... + Gemini per-run'). No internal
contradiction.",
312         "suggestedFix": "",
313         "severity": "low"
314     }
315 ]
316 },
317 {
318     "doc": "docs/ANNOTATE_AND_TRAIN.md",
319     "overallStatus": "significant-staleness",
320     "summary": "A DRAFT tracker whose marketing-page deliverables (P0-P4 plus the
real annotator screenshots P5 mentions) are all shipped and merged (#15/#22/#23/#25), yet the
doc was last touched 2026-06-20 (#23) and never updated to reflect that. The core
engine/annotator technical claims (schema bridge, ship-gate refusing a commercial ship, MIT
annotator v1.6.4, guardrails) all still verify as correct ? only their file:line citations have
drifted. Main staleness: status still DRAFT despite shipped work, P5 marked Todo though
annotator screenshots are committed, the doc is NOT actually indexed in docs/README.md (claimed
three times), and the page since grew 6-language localization the tracker never anticipated. It
is an archive candidate once honestly closed out.",
321     "findings": [
322     {
323         "claim": "Status: DRAFT ? owner-gated on scope (header, line 3); Last
updated: 2026-06-20.",
324         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:3",
325         "verdict": "stale",
326         "evidence": "git log --follow shows the marketing-page work merged: P0 doc
(#15, 6b62ebf), own-model page P1-P4 (#22, b1110e1), org link (#23, 87520b8). Per $7, P0-P4 are
all marked Done ? yet the header still reads DRAFT. Lifecycle (line 4) is DRAFT->IN
PROGRESS->DONE; with P0-P4 shipped this is at least IN PROGRESS.",
327         "suggestedFix": "Flip Status to IN PROGRESS (only owner-gated P5-P9 remain)
and bump Last updated; or, if the owner considers the marketing surface complete, mark DONE and
archive per the lifecycle note.",
328         "severity": "medium"
329     },
330     {
331         "claim": "P5 'Capture + commit real screenshots ... replace placeholders' ?
Status ? (Todo), Gated Owner.",
332         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:76",
333         "verdict": "stale",
334         "evidence": "PR #25 (1906c82) 'show the real annotator screenshots'
committed site/public/img/annotate-enable.png (133712 bytes) and annotate-tool.png (153133
bytes); own-model.html:111-112 references them as the real VLC-UI shots (no people/footage). The
annotator-UI half of P5 is done and merged, not Todo.",
335         "suggestedFix": "Mark P5 ? for the annotator-UI screenshots (cite #25); if
footage screenshots with consented people are still pending, split that into its own row so the
done part isn't hidden under a blanket Todo.",
336         "severity": "high"

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337         },
338         {
339             "claim": "Doc is 'index[ed] in docs/README.md' (line 5), §7 P0 'indexed in
docs/README.md' (? , #15), and DoD 'doc indexed in docs/README.md'.",
340             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:5,71,91",
341             "verdict": "stale",
342             "evidence": "grep -ni 'annotate|own.model|train.your|ANNOTATE_AND'
docs/README.md returns NOT FOUND; the docs/README.md 'Tracked work' list (lines 1-30) does not
mention ANNOTATE_AND_TRAIN.md or own-model. The claimed index entry does not exist.",
343             "suggestedFix": "Either add an ANNOTATE_AND_TRAIN.md entry to
docs/README.md's 'Tracked work' list, or correct P0/DoD to reflect that it is not yet indexed.",
344             "severity": "high"
345         },
346         {
347             "claim": "Engine ship-gate (gate_verdict, harness.py:98) currently refuses a
commercial ship regardless of F1: WASB provenance C3 unresolved + F1@tIoU not calibrated
(#172).",
348             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:12,31",
349             "verdict": "correct",
350             "evidence": "training/gen0/harness.py:177 returns {'ship': False, ...}; :175
'WASB checkpoint provenance C3 UNRESOLVED ? commercial ship blocked regardless of F1'; :174
'ship-gate F1@tIoU target NOT YET CALIBRATED ... see issue #172'. Fact correct; the cited line
(:98) has drifted ? gate_verdict is now at :142.",
351             "suggestedFix": "Update the line citations: gate_verdict is at
harness.py:142 (not :98).",
352             "severity": "low"
353         },
354         {
355             "claim": "Default synth path: synth_trajectory_from_labels uses
csv.DictReader reading row['start_time']/row['end_time'] (stub_runner.py:139) ? MATCHES the
annotator header exactly.",
356             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:46",
357             "verdict": "correct",
358             "evidence": "backend/pipeline/detectors/stub_runner.py:124 def
synth_trajectory_from_labels; :137 for row in _csv.DictReader(f); :139
rallies.append((float(row['start_time']), float(row['end_time']))). Annotator header
rally_number,start_time,end_time,... confirmed at rally-annotator vlc/rally_annotator.lua:128
and docs/CSV_FORMAT.md:6. Claim accurate.",
359             "suggestedFix": "",
360             "severity": "low"
361         },
362         {
363             "claim": "Canonical GT reader load_gt_intervals accepts start/end OR
start_time/end_time (gt_loader.py:14-90).",
364             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:47",
365             "verdict": "correct",
366             "evidence": "backend/eval/gt_loader.py:30 _START_NAMES = ('start',
'start_time'); :44 def load_gt_intervals; :70 headerless positional col0,col1 path. Both aliases
accepted as claimed.",
367             "suggestedFix": "",
368             "severity": "low"
369         },
370         {
371             "claim": "Bundle attests no_paid_llm_training_labels:True (harness.py:192)
and label source is human (harness.py:178).",
372             "docLocation": "docs/ANNOTATE_AND_TRAIN.md:33,60",
373             "verdict": "correct",
374             "evidence": "training/gen0/harness.py:236 'no_paid_llm_training_labels':
True; :222 'label_source': 'human (rally-annotator golden CSVs)'. Facts correct; cited lines
(:192, :178) have drifted to :236 and :222.",
375             "suggestedFix": "Update line citations: no_paid_llm_training_labels at
harness.py:236, label_source at :222.",
376             "severity": "low"
377         },
378         {

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379         "claim": "cmd_train (__main__.py:231) pulls labels/, joins video<->label by
file MD5 + stem-match labels/<name>.rallies.csv <-> videos/<name>.<ext> (__main__.py:271-280).",
380         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:31,49",
381         "verdict": "correct",
382         "evidence": "backend/worker/__main__.py:332 def cmd_train; :359 lists
{corpus}/labels/; :383-384 fname.replace('.rallies.csv','') stem join. Logic present and
correct; cmd_train line drifted from :231 to :332.",
383         "suggestedFix": "Update cmd_train citation to __main__.py:332.",
384         "severity": "low"
385     },
386     {
387         "claim": "Reuse map: add a /own-model alias in site/server.js:49-51; Worker
env.ASSETS.fetch needs no change (site/src/index.js:84).",
388         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52,73",
389         "verdict": "correct",
390         "evidence": "site/server.js:99 'pathname === \"/own-model\" ? \"/own-
model.html\"' and :41 LOCALIZED_PAGES includes own-model.html (alias shifted off the cited
:49-51). site/src/index.js:84 'return env.ASSETS.fetch(request)' with no /own-model branch ?
confirms the Worker needs no change.",
391         "suggestedFix": "Update the server.js alias citation (now ~:99, not
:49-51).",
392         "severity": "low"
393     },
394     {
395         "claim": "Reuse precedent: screenshots under site/public/img/ (precedent
site/og.png + make_og.py).",
396         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:52",
397         "verdict": "correct",
398         "evidence": "git ls-files: site/make_og.py and site/public/og.png both
exist; site/public/img/ holds annotate-enable.png, annotate-tool.png, studio-*.png. The og asset
path is site/public/og.png (the doc wrote 'site/og.png' loosely), make_og.py is at
site/make_og.py.",
399         "suggestedFix": "Minor: the OG image is at site/public/og.png (not
site/og.png).",
400         "severity": "low"
401     },
402     {
403         "claim": "SPA today: web/src/App.tsx is one static screen, no router
(web/package.json); a correction screen needs react-router-dom.",
404         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,64",
405         "verdict": "stale",
406         "evidence": "grep on web/package.json: no react-router dep (claim 'no
router' still TRUE). But App.tsx is no longer 'one static screen' ? it now imports and
conditionally renders SetupWizard (web/src/App.tsx:11,65,242, P2c first-run wizard from PR #88).
The 'one static screen' framing predates the wizard.",
407         "suggestedFix": "Reword to 'no router; App.tsx renders a first-run
SetupWizard + the ingest panel' so the SPA surface description matches the post-#88 code.",
408         "severity": "low"
409     },
410     {
411         "claim": "SPA correction route depends on a new, unbuilt engine endpoint
PATCH /api/segments/{id} (dead columns
candidate_segments.human_label/notes/confirmed_segment_id); P8 ? Gated.",
412         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:53,79",
413         "verdict": "correct",
414         "evidence": "grep across backend/ for 'api/segments'/'PATCH' on
candidate_segments finds only the candidate_segments table + query_candidate_segments readers
(database.py:217, eval/*.py); no PATCH /api/segments route exists. P8 is still genuinely unbuilt
as stated.",
415         "suggestedFix": "",
416         "severity": "low"
417     },
418     {
419         "claim": "Relation note: peer of LOCAL_APPLIANCE_ARCHITECTURE.md (the latter
lands with PR #13).",

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420         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:6,84",
421         "verdict": "stale",
422         "evidence": "git ls-files confirms docs/LOCAL_APPLIANCE_ARCHITECTURE.md
already exists in the repo; it has long since landed, so the forward-looking 'lands with PR #13'
/ 'cross-link once PR #13 merges' phrasing is outdated.",
423         "suggestedFix": "Drop the 'lands with PR #13' future-tense framing ? the doc
exists now; add the cross-link.",
424         "severity": "low"
425     },
426     {
427         "claim": "rally-annotator is a single-file VLC Lua extension, VERSION 1.6.4,
MIT-licensed, public at github.com/Khelsutra/rally-annotator.",
428         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:29",
429         "verdict": "correct",
430         "evidence": "rally-annotator/vlc/rally_annotator.lua:1 '(v1.6.4)'; LICENSE
first line 'MIT License'; own-model.html:109/191 links to https://github.com/Khelsutra/rally-
annotator (org-link migration was PR #23, captured in the doc's own history).",
431         "suggestedFix": "",
432         "severity": "low"
433     },
434     {
435         "claim": "$7 tracker: P1/P2/P3/P4 all ? via '(this PR)'.",
436         "docLocation": "docs/ANNOTATE_AND_TRAIN.md:72-75",
437         "verdict": "correct",
438         "evidence": "own-model.html exists (P1); /own-model alias in server.js:99
(P2); nav cross-link index.html:255 + footer :620 (P3); verbatim annotator content + GitHub link
in own-model.html:108-191 (P4). All shipped ? git history shows these in PR #22 (b1110e1,
'annotate-train P1-P4'), so '(this PR)' should now read '#22'.",
439         "suggestedFix": "Replace the '(this PR)' placeholders in the PR column with
#22 (and #23 for the org-link follow-up) so the tracker cites real merged PRs.",
440         "severity": "medium"
441     }
442 ]
443 },
444 {
445     "doc": "docs/DRY_RUN_CUJ.md",
446     "overallStatus": "minor-staleness",
447     "summary": "The doc's substantive engineering claims are accurate and well-
grounded: the SPA back-half + B-P2 front-half ingest screen exist
(web/src/components/IngestPanel.tsx, web/src/hooks/useIngestJob.ts, #56), the engine bucket-
ingest routing is verified in code, and the three cited engine PRs (#280/#281/#289) plus all
four cited engine tests exist. The staleness is in metadata/framing: the header still says
Status: DRAFT though docs/README.md correctly calls it DELIVERED with DoD $6 boxes 1-3 done, and
the doc repeatedly describes the multilingual workflow as \"en/kn/hi\" (3 langs) when the SPA
now supports 6 (es/da/id added in #87). It is a completed tracker and an archive candidate once
the parked droplet box is closed.",
448     "findings": [
449         {
450             "claim": "Status: DRAFT ? owner-gated . ... Goal for this session.",
451             "docLocation": "docs/DRY_RUN_CUJ.md:3",
452             "verdict": "stale",
453             "evidence": "docs/README.md:4 marks this doc '? DELIVERED (2026-06-21)'; DoD
$6 boxes 1-3 are all checked [x] (DRY_RUN_CUJ.md:47-49) and the work shipped (#53/#55/#56/#57).
A DRAFT/'goal for this session' header no longer matches the completed, delivered state.",
454             "suggestedFix": "Flip the header to Status: DONE / DELIVERED (2026-06-21)
and drop 'Goal for this session'; note the only open box (droplet, $6 box 4) is owner-parked.
Consider archiving once that box closes.",
455             "severity": "medium"
456         },
457         {
458             "claim": "The SPA is 'Fully en/kn/hi (the switcher / ?lang=)' and the
multilingual workflow is en/kn/hi.",
459             "docLocation": "docs/DRY_RUN_CUJ.md:11",
460             "verdict": "stale",
461             "evidence": "web/src/i18n/index.ts:25 ? `export const SUPPORTED_LOCALES =

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[\"en\\\", \"kn\\\", \"hi\\\", \"es\\\", \"da\\\", \"id\\\"]`; es/da/id were added to the SPA in PR #87
(cf59fbf 'add es/da/id to the marketing site (parity with the SPA's 6 languages)'). The SPA now
supports 6 languages, not 3.",
462     "suggestedFix": "Update 'en/kn/hi' to 'en/kn/hi/es/da/id (6 languages)'
where it describes the current switcher; the Hindi/Kannada run results in §7 stay accurate as
historical record.",
463     "severity": "low"
464 },
465 {
466     "claim": "This SPA front-half (ingest/process UI) is the *only* net-new work
for Path B (= B-P2); the SPA has no ingest/process UI today.",
467     "docLocation": "docs/DRY_RUN_CUJ.md:14",
468     "verdict": "correct",
469     "evidence": "B-P2 was net-new work at draft time and has since shipped:
web/src/components/IngestPanel.tsx + web/src/hooks/useIngestJob.ts (first landed in PR #56
'feat(web): B-P2 ? in-UI bucket ingest', 4525c5d, 2026-06-21). useIngestJob.ts drives POST
/api/process (processVideo) ? poll GET /api/jobs/{id} (getJob) ? hand video_id to App, exactly
as scoped.",
470     "suggestedFix": "",
471     "severity": "low"
472 },
473 {
474     "claim": "POST /api/process accepts gs://gcs://s3://local:// URIs and skips
the local-existence check for non-local backends; resolve via storage.resolve_input_ref;
gs://?gcs normalization.",
475     "docLocation": "docs/DRY_RUN_CUJ.md:15,68-72",
476     "verdict": "correct",
477     "evidence": "Engine backend/main.py:581 `input_ref =
storage.resolve_input_ref(req.video_path, ...)` and main.py:590 `if input_ref.backend ==
'local' and not os.path.exists(input_ref.key):` (the non-local skip), with the fetch at
main.py:458 `storage.acquire_local_input(...)`; backend/storage/ref.py:60 `if scheme == 'gs':
scheme = 'gcs'`. Substance matches; the exact line numbers 159/274/283 cited in the doc have
drifted as the engine shipped further PRs.",
478     "suggestedFix": "Optionally refresh the cited line numbers (now
~main.py:581/590/458, ref.py:60) since the engine surface moved; the routing logic itself is
intact.",
479     "severity": "low"
480 },
481 {
482     "claim": "Tested by
tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches,
::test_process_hosted_mode_rejects_nonlocal_uri, ::test_process_unknown_scheme_returns_400,
::test_cli_nonlocal_input_fetches_and_keeps_uri.",
483     "docLocation": "docs/DRY_RUN_CUJ.md:15,76",
484     "verdict": "correct",
485     "evidence": "Engine repo grep: tests/test_api_endpoints.py:508
test_process_nonlocal_uri_skips_exists_and_fetches; :577
test_process_unknown_scheme_returns_400; :608 test_process_hosted_mode_rejects_nonlocal_uri;
:627 test_cli_nonlocal_input_fetches_and_keeps_uri. All four exist with the asserted names.",
486     "suggestedFix": "",
487     "severity": "low"
488 },
489 {
490     "claim": "B-P1 (gs://?/api/process?API-DB video_id) already shipped in
engine M2 / #280; the job-drain hang fixed in #281; the compile-from-bucket 500 fixed in #289.",
491     "docLocation": "docs/DRY_RUN_CUJ.md:15,63,68,88",
492     "verdict": "correct",
493     "evidence": "Engine git log: e111add 'feat(serving): M2 ? URI-aware input
... (#280)'; 41d45fd 'fix(server): drain async jobs under `python -m backend.main` (double-
module hang) (#281)'; e68b583 'fix(api): resolve a bucket-ingested video's source to a local
file before stitching (#289)'. All three PRs exist with descriptions matching the doc.",
494     "suggestedFix": "",
495     "severity": "low"
496 },
497 {

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498         "claim": "DoD §6 box 3: in-UI Path-B bucket?reel CUJ DONE (live e2e
2026-06-21) ? 12 rallies loaded ? reel downloaded; box 4 (CPU droplet) still open/un-
provisioned.",
499         "docLocation": "docs/DRY_RUN_CUJ.md:49-50",
500         "verdict": "correct",
501         "evidence": "Box 3 is corroborated by PR #57 'docs(dry-run): Path-B live e2e
DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3 ?)' and the merged B-P2 SPA (#56).
docs/README.md:6-7 confirms box 4 (droplet, option 3) remains 'parked by the owner (GPU
subscription)'. The run-result specifics (12 rallies, reel sizes, GCS screencast paths) are
runtime artifacts not locally reproducible.",
502         "suggestedFix": "",
503         "severity": "low"
504     },
505     {
506         "claim": "Run-result specifics: 12 rallies / video_id 05e0e577 /
50.9MB+76.5s reels / screencasts at gs://khelsutra-rally-corpus/promo/*.mp4 and the staged
bucket video gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4.",
507         "docLocation": "docs/DRY_RUN_CUJ.md:54-60,82-85",
508         "verdict": "unverifiable",
509         "evidence": "These are GCS bucket objects and a one-off local run's outputs;
they cannot be checked from the repo. No counter-evidence found ? the surrounding code/PR claims
that frame them are all verified.",
510         "suggestedFix": "",
511         "severity": "low"
512     }
513 ]
514 },
515 {
516     "doc": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md",
517     "overallStatus": "significant-staleness",
518     "summary": "A still-active DRAFT tracker (last updated 2026-06-21) whose §9
progress and several \"Absent (net-new)\" engine-fact anchors have drifted: the engine has since
shipped /api/capabilities, exposure-safety boot posture + an edge-auth layer, and merged
native_wasb_runner to master, while the alpha deploy went LIVE on 2026-06-22
(app.khelsutra.guru). The khelsutra-side claims (PR refs #11/#56/#64/#66, IngestPanel,
VITE_API_BASE) check out; the staleness is concentrated in engine status and the deploy/DoD
progress. Not yet an archive candidate (open owner gates remain) but needs a refresh pass.",
519     "findings": [
520     {
521         "claim": "§3 lists `GET /api/capabilities` under 'Absent (net-new): ...
`[verified ? grep returns none]`; §4.6 Reuse map lists it under 'Net-new'; §9 T7 (capabilities
+ health + banner) is ? Todo.",
522         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:49,113,164",
523         "verdict": "stale",
524         "evidence": "engine origin/master backend/main.py:340
`@app.get(\"/api/capabilities\")` (and main.py:307 `/api/health`, main.py:334 `/api/version`)
all exist on master. git grep '/api/capabilities' origin/master returns a hit.",
525         "suggestedFix": "Move `/api/capabilities` out of 'Absent (net-new)' and out
of §4.6 'Net-new'; mark the engine half of T7 (capabilities+health) as shipped (?/?), leaving
only the SPA banner pending.",
526         "severity": "high"
527     },
528     {
529         "claim": "TL;DR + Risk table: 'the engine has zero app-layer auth and CORS
`*` `[verified backend/main.py:42]` ? structural safety only because the engine binds
127.0.0.1.",
530         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:14,30,103,128",
531         "verdict": "stale",
532         "evidence": "engine origin/master backend/main.py:82 uses
`allow_origins=cors_origins_from_env()` (default ['*'] but env-lockable) with
`allow_credentials=False`; main.py:109 adds TrustedHostMiddleware (DNS-rebinding/Host
allowlist); backend/api/auth.py + config/startup.py enforce_startup_checks + M9 `Cf-Access-
Authenticated-User-Email` `owner_id` now exist on master. CORS is no longer a hardcoded bare
`*`, and an app-layer edge-auth integration now exists.",
533         "suggestedFix": "Reword to: CORS default is `*` but now env-configurable

```

```

(allow_credentials off), and the engine now ships a Host-header guard + an edge-auth integration
(Cf-Access trust) + exposure-safety boot posture ? so 'zero app-layer auth' is no longer
literally true. Re-anchor the line (main.py:42 no longer points at CORS).",
534         "severity": "high"
535     },
536     {
537         "claim": "$3 / $4.3: 'Native WASB is in-flight on branch `feat/native-wasb-
runner`: `native_wasb_runner.py` exists `[verified]`'.",
538         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:45,96",
539         "verdict": "stale",
540         "evidence": "engine origin/master now contains
backend/pipeline/detectors/native_wasb_runner.py and tests/test_native_wasb_runner.py (git ls-
tree origin/master). The runner has merged to master; the `feat/native-wasb-runner` branch still
exists but the file is no longer branch-only. (trajectory_hybrid._build_native_runner still
raises NotImplementedError at trajectory_hybrid.py:61 ? that sub-claim is still correct.)",
541         "suggestedFix": "Change 'in-flight on branch feat/native-wasb-runner' to
'native_wasb_runner.py is on master; trajectory_hybrid still refuses, fusion builds it; owner
byte-parity gate still pending.'",
542         "severity": "medium"
543     },
544     {
545         "claim": "$9 alpha-deploy block: 'the actual stand-up (tunnel + Pages + CF
Access + key rotation) is the **owner** step'; $11 DoD box 1 (CUJ-1 e2e on droplet behind gate)
and box 2 (raw :8000 refused, tested) unchecked.",
546         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:175,182,183",
547         "verdict": "stale",
548         "evidence": "deploy/droplet/README.md:100-106 '? Live-verified on the
droplet (2026-06-22)': engine booted on 168.144.126.176; suite 1313 passed; boot-posture fail-
fast, Host-guard 400?200, path-jail, T4 upload?jail, and edge-auth 401-gate all exercised live;
full DNS?Cloudflare?Access?tunnel?engine chain confirmed; app.khelsutra.guru up. Only Gemini-key
rotation + a browser OTP login remain.",
549         "suggestedFix": "Update $9 to note the alpha stand-up is LIVE (2026-06-22,
app.khelsutra.guru, posture test exercised) with only key rotation + browser-OTP login
outstanding; tick/qualify the corresponding DoD boxes.",
550         "severity": "high"
551     },
552     {
553         "claim": "Status header: 'Status: DRAFT ... Last updated: 2026-06-21' with
$9 the source of truth.",
554         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:3",
555         "verdict": "stale",
556         "evidence": "khelsutra git log shows merges through #94 (master HEAD
50b2bd7) and the alpha went live 2026-06-22, both after the doc's 2026-06-21 'last updated'.
Multiple $9 rows and engine facts have moved since (T7 engine half, alpha deploy, native_wasb on
master).",
557         "suggestedFix": "Bump 'Last updated' and consider promoting DRAFT?IN
PROGRESS; refresh $9 rows to current reality.",
558         "severity": "medium"
559     },
560     {
561         "claim": "D4 'Embed RallyPicker ... PR [#11]' and $9 T4 '[#66]', T12 'SPA
#56', alpha-deploy '[#64]' / 'VITE_API_BASE'.",
562         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:33,161,169,175",
563         "verdict": "correct",
564         "evidence": "khelsutra git log: cb74ce5 '...RallyPicker (#11)'; ca619fb 'T4
? in-UI file upload (#66)'; 4525c5d 'B-P2 ? in-UI bucket ingest (#56)'; 8bbc557 'VITE_API_BASE
(#64)'. web/src/components/IngestPanel.tsx handles both chunked upload (T4) and a gs:// URI
(T12); web/src/api/client.ts:25 reads VITE_API_BASE.",
565         "suggestedFix": "",
566         "severity": "low"
567     },
568     {
569         "claim": "T12 (bucket-URI ingest) marked ?: 'engine accepts URIs (M2/#280) +
SPA ingest screen shipped (B-P2, #56); in-UI gs:///reel CUJ proven live e2e (DRY_RUN_CUJ.md
$9)'.",

```

```

570         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:169",
571         "verdict": "correct",
572         "evidence": "docs/DRY_RUN_CUJ.md:80-92 §9 records the live in-UI gs://?reel
CUJ (12 rallies ? reel downloaded) and the compile-from-bucket 500 fix (engine #289);
IngestPanel.tsx accepts a gs:// URI. khelsutra #56 (B-P2) is merged.",
573         "suggestedFix": "",
574         "severity": "low"
575     },
576     {
577         "claim": "T11 retention marked ?: sweep tool shipped (engine #299), auto-
scheduling tracked (engine #301).",
578         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:168,130",
579         "verdict": "unverifiable",
580         "evidence": "These reference the engine repo's GitHub PR #299 / issue #301
(Khelsutra/badminton-highlight-indexer). Not falsifiable from the local khelsutra checkout; the
local engine checkout is on a feature branch and I did not confirm the specific sweep tool by PR
number. The khelsutra-side row metadata is internally consistent (#69 flipped T11 to ? per git
log bea8a2a).",
581         "suggestedFix": "",
582         "severity": "low"
583     },
584     {
585         "claim": "§3 Droplet reality: co-hosts the marketing site at
`/srv/khelsutra-site`, node `:8787`, engine at `~/rally`; MinIO 9000/9001 + GCS emulator 9023.",
586         "docLocation": "docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48",
587         "verdict": "unverifiable",
588         "evidence": "These are live-droplet filesystem/port facts (168.144.126.176)
not checkable from the local repo. deploy/DEPLOY_ALPHA.md:31 and deploy/droplet/README.md:4
corroborate the IP/CPU-no-GPU/blr1 framing, but the exact paths/ports (/srv/khelsutra-site,
:8787, ~/rally) are not in-repo verifiable.",
589         "suggestedFix": "",
590         "severity": "low"
591     },

```

4. user (2026-06-22T09:46:28.188Z)

```

1      # Multilingual Expansion ? 7 new locales (th · id · da · es · zh-Hans · ja · ko)
2
3      > **Status:** `IN PROGRESS` ? **Phase A (es/da/id) shipped
([#67](https://github.com/avidullu/khelsutra/pull/67)); Phases B (Thai) / C (CJK) owner-gated**
· **Owner:** Avi Dullu · **Created:** 2026-06-21 · **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; index in
`docs/README.md`; pointer in memory `current-state`; thread in `NEXT_STEPS.md` once work starts.
6      > **Relation to existing docs:** **PEER-OF / extends**
[`FRIENDLY_AND_MULTILINGUAL.md`](archives/FRIENDLY_AND_MULTILINGUAL.md) (the `en`/`kn`/`hi`
foundation) and the engine design `I18N_PLAN.md`. **Reuses that foundation verbatim**
(react-i18next + i18next-icu + browser detector + key-parity/no-bare-strings CI + self-hosted
OFL Noto via `@fontsource` + Gemini-draft?native-review). This is a **parallel** line: it adds
locales, it does **not** re-architect i18n.
7      > **Honesty note:** each claim tagged `[verified]` (checked vs the working tree this
session), `[researched]` (sourced, needs sign-off), or `[design]` (proposed). Not legal advice ?
the OFL-font carve-out is the owner's standing call (D7 in FRIENDLY).
8
9      ---
10
11     ## 0. TL;DR
12     Add **Thai, Indonesian, Danish, Spanish, Chinese (Simplified), Japanese, Korean** to the
Studio SPA with the **same rigor as en/kn/hi**. The good news `[verified]`: the i18n foundation
already scales ? **zero changes to the negotiation contract or the catalog format**; adding a
locale is "drop a reviewed `common.json` + (if non-Latin) bundle one OFL Noto font + keep CI
key-parity green." The work splits cleanly into **Phase A** (es/da/id ? *catalog-only, no
fonts*), **Phase B** (Thai ? *one small OFL font*), and **Phase C** (zh-Hans/ja/ko ? *large OFL
CJK fonts + lazy-load + an offline-bundle subsetting decision + the zh script-variant call*).

```


****The single most important caveats:**** (1) ****native-speaker review per language**** is the real cost and the public-launch gate (Gemini drafts are **inference-only**, never the source of truth ? the standing guardrail); (2) ****CJK fonts add tens of MB**** to the otherwise-tiny air-gapped bundle; (3) ****Spanish now has a 3-form plural**** (*{one, many, other}*) ? the one authoring wrinkle.

13

14 **## 1. Problem & goal**

15 ****Why now.**** en/kn/hi shipped and proved the foundation is locale-pluggable (a Kannada-default browser auto-renders Kannada; the switcher walks kn?hi?en). The owner wants to broaden reach beyond the India beachhead ? ****SE Asia (th, id), Europe (da, es), and East Asia (zh, ja, ko)****.

16 ****Concrete outcome ("good"):** a user whose browser is in any of the 7 languages lands on the Studio (and, once ``site/`` is localized via FRIENDLY B3, the marketing pages too), and ? without help ? understands the product, runs the tool, and gets a reel, ****entirely in their language****, with the fonts rendering crisply offline.

17 ****Read to get current:** this doc §3?§4;

[`FRIENDLY\_AND\_MULTILINGUAL.md`](archives/FRIENDLY_AND_MULTILINGUAL.md) (the foundation this rides on, esp. §4 design + §6.5 negotiation tests); ``web/src/i18n/index.ts`` (the detector/ICU/fallback config) + ``web/src/main.tsx`` (the ``@fontsource`` font-load pattern) + ``web/src/i18n/hygiene.test.ts`` (key-parity + no-bare-strings + offline guards).

18

19 **## 2. Decisions locked (inherited from FRIENDLY unless noted)**

20 | # | Decision | Source / date | Implication |

21 |---|-----|-----|-----|

22 | ED1 | ****Same framework, no re-architecture**** ? react-i18next + i18next-icu + ``i18next-browser-languagedetector``; one ``en`` source catalog + per-locale siblings; ****always fall back to ``en``**** | FRIENDLY D-* / ``i18n/index.ts`` ``[verified]`` | Each new locale = add to ``SUPPORTED_LOCALES`` + a ``common.json``; negotiation/format unchanged |

23 | ED2 | ****Self-host OFL Noto via ``@fontsource``**** (the D7 OFL-for-media carve-out) ?

****extend it to the 4 new Noto fonts**** (Thai, SC, JP, KR) | FRIENDLY D7 (owner, 2026-06-21) | No new licensing decision; same ``main.tsx`` import + ``unicode-range``-scoped pattern as kn/hi |

24 | ED3 | ****Gemini-draft ? native-speaker review**** (inference only, never training; every value flagged for review) | ``CLAUDE.md`` guardrail | Translation **quality** is human-gated + owner-paced; CI proves parity, not quality |

25 | ED4 | ****CI key-parity + no-bare-strings stay merge gates**** for every locale | ``hygiene.test.ts`` ``[verified]`` | A new catalog must match the ``en`` key set exactly or CI fails |

26 | ED5 | ****All 7 are LTR**** ? no RTL/bidi layout work in this batch | ``[researched]`` CLDR/W3C | No ``dir="rtl"``, no logical-property audit; RTL (ar/he/ur) would be a separate, larger track |

27 | ED6 | ***(proposed)*** | ****Chinese = ``zh-Hans`` (Simplified) for v1****; ``zh-Hant`` (Traditional) is a **separate** catalog + font, added later only on demand | §8 Q2 ``[design]`` | Key the catalog ``zh-Hans`` (not bare ``zh``) so ``zh-Hant`` slots in cleanly |

28

29 **## 3. Foundation ? research / prior art ``[researched]``**

30 Per-language diligence (Unicode CLDR plural-rules chart; SIL OFL; ``@fontsource``; W3C BCP-47 ? sources in §10):

31

32 | Lang | BCP-47 | CLDR ****cardinal**** plurals | Dir | Dedicated font (OFL) | ``@fontsource(-variable)`` pkg | Font size / notes |

33 |---|---|---|---|---|---|

34 | Thai | ``th`` | ****other**** (1 form) | LTR | ****Noto Sans Thai**** | ``noto-sans-thai`` | ****small**** (single script block) · no-space word-wrap gotcha |

35 | Indonesian | ``id`` | ****other**** (1 form) | LTR | ? (Latin) | ? | pure catalog add |

36 | Danish | ``da`` | ****one, other**** (2) | LTR | ? (Latin) | ? | same shape as English (``ø`` ``å`` in Latin font) |

37 | Spanish | ``es`` | ****one, many, other**** (3) ? | LTR | ? (Latin) | ? | ``many`` = millions/compact only; ~15?30% string expansion |

38 | Chinese (Simpl.) | ****zh-Hans**** | ****other**** (1 form) | LTR | ****Noto Sans SC**** | ``noto-sans-sc`` | ****very large**** (~7?16 MB/weight unsubset) · script-variant decision |

39 | Japanese | ``ja`` | ****other**** (1 form) | LTR | ****Noto Sans JP**** | ``noto-sans-jp`` | ****very large**** (CJK) |

40 | Korean | ``ko`` | ****other**** (1 form) | LTR | ****Noto Sans KR**** | ``noto-sans-kr`` | ****very large**** (Hangul + Han) |

41

42 ****Three findings that drive the design:****

43 1. **Spanish's 3-form plural** is the only authoring change. Current CLDR `es` cardinal = `{one, many, other}`; `many` fires only on large/compact numbers (e.g. millions), but every `es` ICU plural message must still carry a `many` arm or `intl-messageformat` falls back. Flag it to the reviewer and extend the parity/lint check. (th/id/zh/ja/ko are **single-form** ? trivial; da is 2-form like en.)

44 2. **Fonts split the batch.** es/da/id need **no new font** (Latin). th needs one **small** OFL font. **zh-Hans/ja/ko** need large OFL CJK fonts ? all OFL-1.1 (bundleable under ED2), all on `@fontsource-variable`, but multi-MB each.

45 3. **CJK + the offline guard collide.** `@fontsource` serves CJK as ~100+ `unicode-range` `woff2` chunks, fetched on demand ? efficient **online**, but the SPA's **air-gapped/offline guarantee** (`hygiene.test.ts` "offline / self-contained") means we must ship **all** chunks so any glyph renders with no network ? **tens of MB** added with 3 CJK languages. Mitigations in §4.

46

47 **## 4. Design / architecture**

48 **Reuse map (what's reused vs new):**

49 - **Reused unchanged:** the negotiation contract (persisted pref ? `?lang=` ? navigator ? `en`), the ICU catalog format, the `@fontsource` `unicode-range`-scoped self-host pattern (main.tsx:7-8), the key-parity + no-bare-strings + offline CI (`hygiene.test.ts`), the draft?review pipeline, the `` sync (`i18n/index.ts:59-65`).

50 - **New:** 7 `common.json` catalogs; `SUPPORTED\_LOCALES` entries (`i18n/index.ts:18`); **4** new Noto font imports; an **es 3-form plural** allowance in the catalog + tooling; a **CJK lazy-load + offline-subset** strategy (Phase C); Thai line-break CSS safety net.

51 - **?** Catalog-home (resolved by FRIENDLY B3): catalogs now live at repo-root **~/locales/{en,kn,hi,es,da,id}/common.json** ? the shared top-level dir from the I18N_PLAN "D-CATALOG-HOME" diagram. FRIENDLY **B3** hoisted them out of `web/src/i18n/locales/` (the SPA re-points its imports there; the `site/` shim consumes a `copy:locales` mirror). These es/da/id catalogs moved with the hoist, as anticipated.

52

53 **Negotiation tests (per FRIENDLY §6.5):** every new locale gets an N1/N2-style test ? a browser set to that language **auto-renders** it (e.g. `navigator.languages=["ja-JP"]` ? `ja`), region subtags normalize (`zh-Hans-CN` ? `zh-Hans`), and a missing key falls back to `en`. **Auto-selection is the promise; it's proven by a test, never assumed.**

54

55 **Phasing (least new code/assets first):**

56 - **Phase A ? Latin trio (`es`, `da`, `id`):** catalog-only, zero new assets. **Highest ROI**, proves the pipeline scales beyond Indic with no code. Only `es` needs the 3-form-plural touch.

57 - **Phase B ? Thai (`th`):** one small OFL font. **Reuses** the kn/hi font pattern exactly; add the no-space word-wrap CSS safety net. Self-contained, low-risk.

58 - **Phase C ? CJK (`zh-Hans`, `ja`, `ko`):** the heavy slice. **Needs** the zh-Hans/Hant decision, 3 large OFL fonts, **per-locale lazy font-load** (so non-CJK users never download CJK), and an **offline-bundle subset strategy** (ship-all-chunks vs a custom subset build step vs online-lazy-only). One PR per CJK language.

59

60 **## 5. Privacy / scope note**

61 i18n is **presentation-only** ? no new personal-data collection, no biometrics, minors-excluded posture unchanged (same as FRIENDLY §5). Gemini-draft translation is inference, not training, with no user data in the loop. CJK/Thai font bundling is OFL media, not code/weights.

62

63 **## 6. Honest limits ? what this does NOT do**

64 - A green CI / DONE proves **key-parity + render + no-bare-strings + auto-negotiation**, **not** translation **quality** (native-review-gated, ED3) or correct font rendering (owner-eyeball-gated, like kn/hi).

65 - **zh-Hant** (Traditional) is out of v1 (separate catalog + font); only `zh-Hans` ships first.

66 - **No RTL** ? this batch is all LTR.

67 - The **CJK offline bundle is large** (tens of MB) unless a subset build is adopted ? a real size trade-off, not free.

68 - `site/` marketing localization for these 7 **depends on FRIENDLY B3** landing first (the shared shim); until then this is SPA-only.

69

70 **## 7. Deliverables & progress tracker** ? **source of truth**

71 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking.

```

72
73 | ID | Deliverable | Phase | Depends on | Gated? | Status | PR |
74 |----|-----|-----|-----|-----|-----|----|
75 | E0 | This proposal doc + inbound pointers (README/NEXT_STEPS/memory) | ? | ? | **Owner
approval to start** | ? | [#58](https://github.com/avidullu/khelsutra/pull/58) (owner-merged) |
76 | EA1 | **Spanish (`es`)** catalog (Gemini-draft) + ICU **3-form `{one,many,other}`**
handling + extend key-parity/no-bare-strings tooling for the `many` arm + negotiation test | A |
E0 | No (after approval) | ? | [#67](https://github.com/avidullu/khelsutra/pull/67) |
77 | EA2 | **Danish (`da`)** catalog + negotiation test | A | E0 | No | ? |
[#67](https://github.com/avidullu/khelsutra/pull/67) |
78 | EA3 | **Indonesian (`id`)** catalog + negotiation test | A | E0 | No | ? |
[#67](https://github.com/avidullu/khelsutra/pull/67) |
79 | EB1 | **Noto Sans Thai** bundle (OFL, `@fontsource-variable`) + Thai no-space word-
wrap CSS safety net | B | E0 | No (ED2 carve-out applies) | ? | ? |
80 | EB2 | **Thai (`th`)** catalog ? render + negotiation test | B | EB1 | No | ? | ? |
81 | EC0 | **Decision + design:** `zh-Hans` vs `zh-Hant` (ED6) + CJK **lazy per-locale
font-load** + **offline-bundle subset** strategy (ship-all-chunks vs custom subset vs online-
lazy) | C | E0 | **Owner (Q2/Q3)** | ? | ? |
82 | EC1 | **Noto Sans SC** + **zh-Hans** catalog + lazy-load + negotiation test | C |
EC0 | No (after EC0) | ? | ? |
83 | EC2 | **Noto Sans JP** + **ja** catalog + lazy-load + negotiation test | C | EC0 |
No | ? | ? |
84 | EC3 | **Noto Sans KR** + **ko** catalog + lazy-load + negotiation test | C | EC0 |
No | ? | ? |
85 | EX1 | **Native-speaker review** of each AI-drafted catalog (one tracked row; per-
language sign-off) | ? | the catalog rows | **Owner (reviewer sourcing)** | ? | ? |
86 | EX2 | **Owner-machine font eyeball** per new script (th, zh, ja, ko) ? cloud CI can't
render fonts | ? | EB*/EC* | **Owner (owner machine)** | ? | ? |
87 | EX3 | **`site/` marketing parity** for the 7 (rides FRIENDLY **B3** shared `t()` shim)
| ? | FRIENDLY B3 | Blocked on B3 | ? | ? |
88
89 ## 8. Open questions ? owner
90 1. **Approve starting this line** (in parallel with the FRIENDLY UX/A-items), and the
**A ? B ? C phasing** ? (E0 gate.)*
91 2. **Chinese:** ship **zh-Hans` only** for v1 (recommended), or **both Hans + Hant** ?
*(EC0; Hant = a whole extra catalog + Noto Sans TC.)*
92 3. **CJK offline strategy** *(EC0; the real size trade-off):* (a) ship **all** font
chunks ? simplest, **tens of MB** to the air-gapped bundle; (b) a **custom subset build**
restricted to glyphs the reviewed catalogs use ? KB-scale but adds a build step that re-runs on
catalog change; (c) **online lazy-load only** ? small bundle but **breaks the offline guarantee
for CJK**. *Recommend (b) if offline-for-CJK matters, else (a).*
93 4. **Native-reviewer sourcing** for 7 languages ? who/how/budget? (Vendor like the Roora
labeling pilot? community? paid freelancers?) This is the launch-gate cost.
94 5. **Priority order** ? which markets first? (Default A-then-B-then-C is by *engineering
cost*; if a market is urgent, e.g. `id` or `ja`, it can jump within its phase.)
95 6. **Does `site/` localize for these too** for v1, or SPA-only until FRIENDLY B3 lands?
*(EX3.)*
96
97 ## 9. Definition of done
98 - [ ] All 7 catalogs (`es`, `da`, `id`, `th`, `zh-Hans`, `ja`, `ko`) live +
**rendering** on the `web/` SPA; switcher + persistence + `en` fallback; **no raw keys**.
99 - [ ] **Auto-negotiation green** ? a browser set to each language auto-renders it
(N1/N2-style tests per locale); region subtags normalize; missing-key ? `en`.
100 - [ ] **CI key-parity + no-bare-strings green** (incl. the `es` 3-form plural) + the
**offline/self-contained guard** still green after the CJK fonts land.
101 - [ ] **Fonts render correctly** ? owner eyeball per new script (th, zh-Hans, ja, ko).
102 - [ ] **Native-speaker review** complete + signed off per language (EX1).
103 - [ ] CJK **offline/bundle-size** strategy implemented per the EC0 decision; documented
bundle-size impact.
104 - [ ] *(if in scope)* `site/` marketing renders the 7 via the shared shim (EX3, after
FRIENDLY B3).
105 - [ ] $7 all rows ? (or explicitly deferred) ? flip **Status ? DONE** + archive.
106
107 ## 10. References
108 **Internal `[verified]`:** `web/src/i18n/{index.ts,hygiene.test.ts}`, `web/src/main.tsx`

```

```
(font-load pattern), repo-root `~/locales/{en,kn,hi,es,da,id}/common.json` (hoisted by FRIENDLY
B3 ? D-CATALOG-HOME); [ `FRIENDLY_AND_MULTILINGUAL.md` ](archives/FRIENDLY_AND_MULTILINGUAL.md),
engine `docs/I18N_PLAN.md`, `docs/UI_PROPOSAL.md`.
109    **External [researched]:** Unicode CLDR v48 plural-rules chart
(https://www.unicode.org/cldr/charts/48/supplemental/language_plural_rules.html) + plural-rules
spec; W3C BCP-47 / Chinese script subtags (`zh-Hans`/`zh-Hant`); SIL Open Font License 1.1 +
Noto CJK LICENSE (github.com/notofonts/noto-cjk); `@fontsource-variable/noto-
sans-{thai,sc,jp,kr}`; W3C Thai line-breaking (`thai-gap`); react-i18next + i18next-icu.
110
111    ---
112
113    ### Changelog
114    - `2026-06-21` ? **Phase A shipped ?
[#67](https://github.com/avidullu/kheIsutra/pull/67)** (E0/EA1/EA2/EA3 ?). es/da/id catalogs
(AI-draft, native review pending) + `lang.*` autonyms + `SUPPORTED_LOCALES`/`LanguageSwitcher`
wiring; **es 3-form {one,many,other}** plural; render + auto-negotiation + plural tests. Added
a **placeholder-parity** hygiene guard (every locale keeps en's `{placeholder}` set) which
caught + fixed a real kn/hi A1-drift (stale `{videoId}` in `mode.loading`/`mode.error`). 168 web
tests green; browser-verified es-da render with all 6 languages in the switcher. Status `DRAFT ?
IN PROGRESS`. Phases B/C still owner-gated.
115    - `2026-06-21` ? created. Owner ask: add
Thai/Indonesian/Danish/Spanish/Chinese/Japanese/Korean with en/kn/hi-level diligence, as a
parallel line for approval. Grounded in a per-language i18n research pass (CLDR plurals, OFL
Noto fonts, BCP-47, offline-bundle sizing) + the verified working-tree foundation. Phasing A
(Latin) ? B (Thai) ? C (CJK); decisions ED1?ED6; 6 owner questions surfaced (E0 = approval
gate).
116
```

5. assistant (2026-06-22T09:46:32.358Z)

Now let me find the staleFindings for my doc. Let me read the rest of the JSON to locate the `staleFindings` array and the perDoc entry for MULTILINGUAL_EXPANSION.md.

6. user (2026-06-22T09:46:34.906Z)

```
603:    "doc": "docs/MULTILINGUAL_EXPANSION.md",
609:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:3,75-78",
617:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:51,108",
625:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:3,79-84",
633:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:87,16,68",
641:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:108",
649:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:114",
657:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:50",
665:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49",
673:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49,50",
681:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:12,43,114",
689:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:6,17,108",
697:    "docLocation": "docs/MULTILINGUAL_EXPANSION.md:5",
699:    "evidence": "docs/README.md:16 '- **`MULTILINGUAL_EXPANSION.md`** ? proposal to
add **7 more locales** (th/id/da/es/zh-Hans/ja/ko)'.",
906:[Omitted long matching line]
909:    "claim": "MULTILINGUAL_EXPANSION.md is a 'proposal to add 7 more locales
(th/id/da/es/zh-Hans/ja/ko) ... DRAFT ? owner-gated (approval to start), [#58].'",
912:    "evidence": "docs/MULTILINGUAL_EXPANSION.md:3 status header reads 'IN PROGRESS ?
Phase A (es/da/id) shipped (#67); Phases B (Thai) / C (CJK) owner-gated'. Git confirms PR #67
'feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs'. site/public/locales/
now contains es, da, id directories and web/src/i18n/index.ts:25 lists SUPPORTED_LOCALES =
en,kn,hi,es,da,id. So es/da/id are no longer 'to add', and the doc is not a not-yet-started
DRAFT.",
913:    "suggestedFix": "Update to: '**`MULTILINGUAL_EXPANSION.md`** ? `IN PROGRESS`:
Phase A (es/da/id) shipped ([#67]); Thai (B) / CJK (C) remain owner-gated. Adds locales beyond
the en/kn/hi foundation; peer of the archived `FRIENDLY_AND_MULTILINGUAL.md`.'",
917:    "claim": "The index lists every tracked doc in this repo (DRY_RUN_CUJ,
FRIENDLY_AND_MULTILINGUAL, MULTILINGUAL_EXPANSION, LOCAL_APPLIANCE_ARCHITECTURE,
PER_RALLY_SELECTION, REQUEST_ACCESS_FORM, PORTABILITY).",
```

```

1123:      "claim": "Doc is archived at docs/archives/ with inbound pointers
(NEXT_STEPS.md, docs/README.md, docs/MULTILINGUAL_EXPANSION.md) updated to the archived path
(#93 changelog).",
1126:      "evidence": "git ls-files: docs/archives/FRIENDLY_AND_MULTILINGUAL.md.
docs/README.md:9 and NEXT_STEPS.md:21 and docs/MULTILINGUAL_EXPANSION.md:6,17,108 all reference
archives/FRIENDLY_AND_MULTILINGUAL.md. Archive merge commit e2570e4 (#93) present.",
1487:      "doc": "docs/MULTILINGUAL_EXPANSION.md",
1489:      "docLocation": "docs/MULTILINGUAL_EXPANSION.md:87,16,68",
1496:      "doc": "docs/MULTILINGUAL_EXPANSION.md",
1498:      "docLocation": "docs/MULTILINGUAL_EXPANSION.md:108",
1505:      "doc": "docs/MULTILINGUAL_EXPANSION.md",
1507:      "docLocation": "docs/MULTILINGUAL_EXPANSION.md:50",
1514:      "doc": "docs/MULTILINGUAL_EXPANSION.md",
1516:      "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49",
1596:      "claim": "MULTILINGUAL_EXPANSION.md is a 'proposal to add 7 more locales
(th/id/da/es/zh-Hans/ja/ko) ... DRAFT ? owner-gated (approval to start), [#58].'",
1599:      "evidence": "docs/MULTILINGUAL_EXPANSION.md:3 status header reads 'IN PROGRESS ?
Phase A (es/da/id) shipped (#67); Phases B (Thai) / C (CJK) owner-gated'. Git confirms PR #67
'feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs'. site/public/locales/
now contains es, da, id directories and web/src/i18n/index.ts:25 lists SUPPORTED_LOCALES =
en, kn, hi, es, da, id. So es/da/id are no longer 'to add', and the doc is not a not-yet-started
DRAFT.",
1600:      "suggestedFix": "Update to: `***`MULTILINGUAL_EXPANSION.md`***` ? `IN PROGRESS`: Phase
A (es/da/id) shipped ([#67]); Thai (B) / CJK (C) remain owner-gated. Adds locales beyond the
en/kn/hi foundation; peer of the archived `FRIENDLY_AND_MULTILINGUAL.md`.'",
1605:      "claim": "The index lists every tracked doc in this repo (DRY_RUN_CUJ,
FRIENDLY_AND_MULTILINGUAL, MULTILINGUAL_EXPANSION, LOCAL_APPLIANCE_ARCHITECTURE,
PER_RALLY_SELECTION, REQUEST_ACCESS_FORM, PORTABILITY).",

```

7. assistant (2026-06-22T09:46:37.408Z)

Let me read the full perDoc findings for my doc (lines 603-720) and the staleFindings entries (1485-1530).

8. user (2026-06-22T09:46:40.309Z)

```

603      "doc": "docs/MULTILINGUAL_EXPANSION.md",
604      "overallStatus": "minor-staleness",
605      "summary": "An accurate, still-live tracker: its core claims hold up ? Phase A
(es/da/id) shipped exactly as described (#67), the locale catalogs live at the stated repo-root
/locales path, and Phases B (Thai) / C (CJK) genuinely have not started (no th/zh/ja/ko catalogs
or fonts in the tree). The staleness is minor and concentrated in two places: (1) EX3 and the
repeated \"SPA-only until FRIENDLY B3\" framing are now outdated ? B3 (#70) and site es/da/id
parity (#87) both shipped, so the marketing site is no longer SPA-only for the Phase-A trio; (2)
a broken §10 reference to a non-existent docs/UI_PROPOSAL.md, plus small line-number/test-count
drift. Not an archive candidate: Phases B/C are real open work.",
606      "findings": [
607      {
608          "claim": "Phase A (es/da/id) shipped via #67; E0/EA1/EA2/EA3 ? (E0 via
owner-merged #58).",
609          "docLocation": "docs/MULTILINGUAL_EXPANSION.md:3,75-78",
610          "verdict": "correct",
611          "evidence": "git log: a2292ef 'feat(web): MULTILINGUAL Phase A ?
Spanish/Danish/Indonesian SPA catalogs (#67)'; a20002f 'docs: propose Multilingual Expansion ?
(#58)'. locales/{es,da,id}/common.json all tracked.",
612          "suggestedFix": "",
613          "severity": "low"
614      },
615      {
616          "claim": "Catalogs live at repo-root
/locales/{en,kn,hi,es,da,id}/common.json (hoisted by FRIENDLY B3, D-CATALOG-HOME).",
617          "docLocation": "docs/MULTILINGUAL_EXPANSION.md:51,108",
618          "verdict": "correct",
619          "evidence": "git ls-files locales/ returns exactly en,kn,hi,es,da,id
common.json; web/src/i18n/index.ts:16-21 imports them from

```

```

'../../../locales/<loc>/common.json'.",
620         "suggestedFix": "",
621         "severity": "low"
622     },
623     {
624         "claim": "Phases B (Thai) and C (zh-Hans/ja/ko) are owner-gated and not
started (EB*, EC* all ?).",
625         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:3,79-84",
626         "verdict": "correct",
627         "evidence": "git ls-files | grep 'locales/(th|zh|ja|ko)' returns nothing
(exit 1); grep for noto-sans-(thai|sc|jp|kr) in web/site package.json returns nothing (exit
1).",
628         "suggestedFix": "",
629         "severity": "low"
630     },
631     {
632         "claim": "EX3 ('site/ marketing parity for the 7') is ? and 'Blocked on B3';
and $1/$4/$6 say site localization 'depends on FRIENDLY B3 landing first ? until then this is
SPA-only.'",
633         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:87,16,68",
634         "verdict": "stale",
635         "evidence": "FRIENDLY B3 shipped: git log 1b1fa7b 'feat(web+site): B3 ?
trilingual marketing site via shared t()/data-i18n shim (#70)'. Site es/da/id parity also
shipped: cf59fbf '#87'. git ls-files shows site/public/locales/{es,da,id}/common.json present.
The dependency is satisfied and the Phase-A trio is no longer SPA-only.",
636         "suggestedFix": "Update EX3 to reflect that FRIENDLY B3 landed (#70) and the
Phase-A es/da/id site parity shipped (#87) ? mark the es/da/id portion ? (site has those
catalogs) with only th/zh/ja/ko site parity remaining; drop the 'until then this is SPA-only'
caveat for es/da/id in $1/$4/$6.",
637         "severity": "medium"
638     },
639     {
640         "claim": "$10 References cites internal 'docs/UI_PROPOSAL.md'.",
641         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:108",
642         "verdict": "stale",
643         "evidence": "git ls-files | grep -i UI_PROPOSAL returns nothing (exit 1) ?
no such file exists anywhere in the repo.",
644         "suggestedFix": "Remove the docs/UI_PROPOSAL.md reference or repoint it to
the correct doc (the engine repo has UI docs; this repo does not contain UI_PROPOSAL.md).",
645         "severity": "low"
646     },
647     {
648         "claim": "Changelog: '168 web tests green' (as of #67, 2026-06-21).",
649         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:114",
650         "verdict": "correct",
651         "evidence": "That was the count at the #67 changelog timestamp and is framed
as a point-in-time changelog note. The live suite now reports 223 passed (24 files) via `npx
vitest run` ? higher because of later PRs (#68/#70/#73/#80/#88 etc.), so the historical 168
figure is not wrong, just superseded.",
652         "suggestedFix": "Optional: the changelog is dated and historical, so no fix
needed; if desired, note the current count is higher.",
653         "severity": "low"
654     },
655     {
656         "claim": "SUPPORTED_LOCALES entries are at i18n/index.ts:18.",
657         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:50",
658         "verdict": "stale",
659         "evidence": "web/src/i18n/index.ts: SUPPORTED_LOCALES is declared at line 25
(= ['en','kn','hi','es','da','id']), not line 18 (line 18 is the hiCommon import). Line-number
drift only.",
660         "suggestedFix": "Change the citation from i18n/index.ts:18 to
i18n/index.ts:25.",
661         "severity": "low"
662     },
663     {

```

```

664         "claim": "The <html lang> sync is at i18n/index.ts:59-65.",
665         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49",
666         "verdict": "stale",
667         "evidence": "web/src/i18n/index.ts:59-65 is the detection-order block
(order/lookupLocalStorage/etc.); the syncDocumentLang() function for <html lang> begins at line
70.",
668         "suggestedFix": "Repoint the <html lang> citation to ~i18n/index.ts:70
(syncDocumentLang).",
669         "severity": "low"
670     },
671     {
672         "claim": "main.tsx:7-8 is the @fontsource unicode-range-scoped font-load
pattern reused for new fonts.",
673         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49,50",
674         "verdict": "correct",
675         "evidence": "web/src/main.tsx:7-8 = import '@fontsource-variable/ noto-sans-
kannada/index.css'; import '@fontsource-variable/ noto-sans-devanagari/index.css'; comment at
lines 5-6 describes the unicode-range-scoped self-host.",
676         "suggestedFix": "",
677         "severity": "low"
678     },
679     {
680         "claim": "es needs a 3-form {one, many, other} ICU plural and the tooling
enforces placeholder parity.",
681         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:12,43,114",
682         "verdict": "correct",
683         "evidence": "locales/es/common.json:51 '{count, plural, one {# jugada} many
{# jugadas} other {# jugadas}}' (and lines 59,466). Placeholder-parity guard exists:
web/src/i18n/hygiene.test.ts:53 'describe(\"placeholder parity ? every locale preserves en's
{placeholders}\\\"'.",
684         "suggestedFix": "",
685         "severity": "low"
686     },
687     {
688         "claim": "The doc is a PEER-OF FRIENDLY_AND_MULTILINGUAL.md, which lives in
docs/archives/.",
689         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:6,17,108",
690         "verdict": "correct",
691         "evidence": "git ls-files shows docs/archives/FRIENDLY_AND_MULTILINGUAL.md
exists (archived via #93 e2570e4). The relative link 'archives/FRIENDLY_AND_MULTILINGUAL.md'
from docs/ resolves correctly.",
692         "suggestedFix": "",
693         "severity": "low"
694     },
695     {
696         "claim": "Doc is indexed in docs/README.md.",
697         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:5",
698         "verdict": "correct",
699         "evidence": "docs/README.md:16 '- **`MULTILINGUAL_EXPANSION.md`** ? proposal
to add **7 more locales** (th/id/da/es/zh-Hans/ja/ko)'.",
700         "suggestedFix": "",
701         "severity": "low"
702     }
703 ]
704 },
705 {
706     "doc": "docs/PER_RALLY_SELECTION.md",
707     "overallStatus": "significant-staleness",
708     "summary": "The doc's MVP thesis and SPA-side deliverables (P0-P5) are accurate
and genuinely shipped, and the future P7/P9/P10 engine endpoints are correctly marked not-built.
But the engine moved on: two whole engine-side items the doc lists as not-done are now shipped
(P6's /api/health AND compile's download_url; P8's /api/videos + db.list_videos), the central
\"/api/compile returns a filepath NOT a URL\" claim is now false, the P3/P4/P5 PR numbers were
never backfilled from \"(this PR)\", and essentially every [verified backend/main.py:NNN] /
database.py line anchor has drifted off its target. Not an archive candidate ? the doc itself is

```

```

still live for the owner-gated M1-M5 work ? but its §4.3 table, §7 tracker, and Foundation
anchors need a refresh pass.",
709         "findings": [
710             {
711                 "claim": "P6 (engine repo) ? GET /api/health + offline banner; /api/compile
returns download_url ? Status ? (Todo, not built); §4.3 table marks GET /api/health as 'new' and
the download_url addition as 'modify' (not yet done).",
712                 "docLocation": "docs/PER_RALLY_SELECTION.md:133 (P7 tracker row P6), :83-84
(§4.3 table)",
713                 "verdict": "stale",
714                 "evidence": "backend/main.py:307 `@app.get(\"/api/health\")` is fully
implemented; backend/main.py compile (~line 769) returns
`{\"status\": \"success\", \"filepath\": \"out_path\", \"download_url\": \"download_url\"}` where
download_url = `/api/highlights/{video_id}/{out_name}`. Both P6 items are shipped in the
engine.",
715                 "suggestedFix": "Flip P6 to ? (or split: health=?, download_url=?) and
update §4.3 status from 'new'/'modify' to 'exists'. Note in the doc that the engine now returns
download_url so the SPA no longer needs to hand-build the URL (though it still can).",
716                 "severity": "high"
717             },
718             {
719                 "claim": "Compile returns a server-local filepath, NOT a URL, so the SPA
constructs this download URL itself (repeated as a core caveat in §0 TL;DR, §3 Foundation, §4.2
data flow, D11/risk table, changelog).",
720                 "docLocation": "docs/PER_RALLY_SELECTION.md:12, :44, :73, :116",
721                 "verdict": "stale",
722                 "evidence": "backend/main.py compile endpoint now returns `download_url`
alongside `filepath`: `download_url =
f\"/api/highlights/{quote(req.video_id)}/{quote(out_name)}\"; return
{\"status\": \"success\", \"filepath\": \"out_path\", \"download_url\": \"download_url\"}`. The 'filepath not
a URL' premise that the whole 'SPA must build the URL' design rests on is no longer the full
picture.",

```

9. user (2026-06-22T09:46:41.303Z)

```

1485     },
1486     {
1487         "doc": "docs/MULTILINGUAL_EXPANSION.md",
1488         "claim": "EX3 ('site/ marketing parity for the 7') is ? and 'Blocked on B3'; and
§1/§4/§6 say site localization 'depends on FRIENDLY B3 landing first ? until then this is SPA-
only.'",
1489         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:87,16,68",
1490         "verdict": "stale",
1491         "evidence": "FRIENDLY B3 shipped: git log 1b1fa7b 'feat(web+site): B3 ?
trilingual marketing site via shared t()/data-i18n shim (#70)'. Site es/da/id parity also
shipped: cf59fbf '#87'. git ls-files shows site/public/locales/{es,da,id}/common.json present.
The dependency is satisfied and the Phase-A trio is no longer SPA-only.",
1492         "suggestedFix": "Update EX3 to reflect that FRIENDLY B3 landed (#70) and the
Phase-A es/da/id site parity shipped (#87) ? mark the es/da/id portion ? (site has those
catalogs) with only th/zh/ja/ko site parity remaining; drop the 'until then this is SPA-only'
caveat for es/da/id in §1/§4/§6.",
1493         "severity": "medium"
1494     },
1495     {
1496         "doc": "docs/MULTILINGUAL_EXPANSION.md",
1497         "claim": "§10 References cites internal 'docs/UI_PROPOSAL.md'.",
1498         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:108",
1499         "verdict": "stale",
1500         "evidence": "git ls-files | grep -i UI_PROPOSAL returns nothing (exit 1) ? no
such file exists anywhere in the repo.",
1501         "suggestedFix": "Remove the docs/UI_PROPOSAL.md reference or repoint it to the
correct doc (the engine repo has UI docs; this repo does not contain UI_PROPOSAL.md).",
1502         "severity": "low"
1503     },
1504 
```



```

1505         "doc": "docs/MULTILINGUAL_EXPANSION.md",
1506         "claim": "SUPPORTED_LOCALES entries are at i18n/index.ts:18.",
1507         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:50",
1508         "verdict": "stale",
1509         "evidence": "web/src/i18n/index.ts: SUPPORTED_LOCALES is declared at line 25 (=
['en','kn','hi','es','da','id']), not line 18 (line 18 is the hiCommon import). Line-number
drift only.",
1510         "suggestedFix": "Change the citation from i18n/index.ts:18 to
i18n/index.ts:25.",
1511         "severity": "low"
1512     },
1513     {
1514         "doc": "docs/MULTILINGUAL_EXPANSION.md",
1515         "claim": "The <html lang> sync is at i18n/index.ts:59-65.",
1516         "docLocation": "docs/MULTILINGUAL_EXPANSION.md:49",
1517         "verdict": "stale",
1518         "evidence": "web/src/i18n/index.ts:59-65 is the detection-order block
(order/lookupLocalStorage/etc.); the syncDocumentLang() function for <html lang> begins at line
70.",
1519         "suggestedFix": "Repoint the <html lang> citation to ~i18n/index.ts:70
(syncDocumentLang).",
1520         "severity": "low"
1521     },
1522     {
1523         "doc": "docs/PER_RALLY_SELECTION.md",
1524         "claim": "P6 (engine repo) ? GET /api/health + offline banner; /api/compile
returns download_url ? Status ? (Todo, not built); $4.3 table marks GET /api/health as 'new' and
the download_url addition as 'modify' (not yet done).",
1525         "docLocation": "docs/PER_RALLY_SELECTION.md:133 (P7 tracker row P6), :83-84
($4.3 table)",
1526         "verdict": "stale",
1527         "evidence": "backend/main.py:307 `@app.get(\"/api/health\")` is fully
implemented; backend/main.py compile (~line 769) returns
`{\"status\":\"success\", \"filepath\":\"out_path\", \"download_url\":\"download_url\"}` where
download_url = `/api/highlights/{video_id}/{out_name}`. Both P6 items are shipped in the
engine.",
1528         "suggestedFix": "Flip P6 to ? (or split: health=?, download_url=?) and update
$4.3 status from 'new'/'modify' to 'exists'. Note in the doc that the engine now returns
download_url so the SPA no longer needs to hand-build the URL (though it still can).",
1529         "severity": "high"

```

10. assistant (2026-06-22T09:46:47.392Z)

I now have the complete worklist for my doc. The staleFindings (verdict === "stale") for `docs/MULTILINGUAL\_EXPANSION.md` are:

1. ****EX3 / "SPA-only until B3"**** (medium) ? B3 (#70) and site es/da/id parity (#87) shipped; lines 87, 16, 68.
2. ****\$10 broken UI_PROPOSAL.md ref**** (low) ? line 108.
3. ****SUPPORTED_LOCALES line drift**** (low) ? `i18n/index.ts:18` ? `:25`; line 50.
4. ****`<html lang>` sync line drift**** (low) ? `i18n/index.ts:59-65` ? `:70`; line 49.

Let me re-verify each against the live repos before editing. First the khelsutra-side facts (git log for #70/#87, the i18n/index.ts line numbers, site/public/locales).

11. user (2026-06-22T09:46:49.117Z)

cf59fbf feat(site): add es/da/id to the marketing site (parity with the SPA's 6 languages) (#87)
1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-HOME) (#70)

12. user (2026-06-22T09:46:50.121Z)

site/public/locales/da/common.json

```
site/public/locales/en/common.json
site/public/locales/es/common.json
site/public/locales/hi/common.json
site/public/locales/id/common.json
site/public/locales/kn/common.json
```

13. user (2026-06-22T09:46:50.876Z)

```
---exit 0---
```

14. assistant (2026-06-22T09:46:53.830Z)

Confirmed: #70 and #87 both shipped; site/public/locales has en, kn, hi, es, da, id (the Phase-A trio present). UI_PROPOSAL.md does not exist in khelsutra. Now let me verify the i18n/index.ts line numbers.

15. user (2026-06-22T09:46:54.353Z)

```
1      // i18n scaffold (tracker B1) ? the shared localization foundation for KhelSutra Studio.
2      //
3      // Per docs/archives/FRIENDLY_AND_MULTILINGUAL.md + the engine's I18N_PLAN.md:
4      //   - keys, never prose (t("app.title"), enforced later by a no-bare-strings lint, B2);
5      //   - ICU MessageFormat for interpolation/plurals (Kannada/Hindi plural rules live in
the catalog,
6      //     not the code ? replaces the SPA's hand-rolled `count > 1 ? "s" : ""`);
7      //   - one negotiation order shared with the marketing site (B3);
8      //   - always-fallback to `en` so a missing translation degrades to English, never a raw
key.
9      import i18n from "i18next";
10     import { initReactI18next } from "react-i18next";
11     import LanguageDetector from "i18next-browser-languagedetector";
12     import ICU from "i18next-icu";
13     // Catalogs are HOISTED to repo-root /locales (D-CATALOG-HOME) ? the single source of
truth shared
14     // by this SPA and the marketing site's t() shim (tracker B3). Vite resolves the
workspace root via
15     // the repo `.git` marker, so importing the parent-dir /locales works in dev, build, and
vittest.
16     import enCommon from "../../locales/en/common.json";
17     import knCommon from "../../locales/kn/common.json";
18     import hiCommon from "../../locales/hi/common.json";
19     import esCommon from "../../locales/es/common.json";
20     import daCommon from "../../locales/da/common.json";
21     import idCommon from "../../locales/id/common.json";
22
23     // Supported locales: en/kn/hi foundation (FRIENDLY D6) + MULTILINGUAL_EXPANSION Phase A
(es/da/id,
24     // Latin, catalog-only). Anything else falls back to `en`.
25     export const SUPPORTED_LOCALES = ["en", "kn", "hi", "es", "da", "id"] as const;
26     export type Locale = (typeof SUPPORTED_LOCALES)[number];
27     export const DEFAULT_LOCALE: Locale = "en";
28
29     // ONE persisted-preference key, shared with the marketing site's t() shim (D9 / tracker
B3), so a
30     // visitor's language choice carries across both surfaces.
31     export const LANG_STORAGE_KEY = "khelsutra.lang";
32
33     // Catalogs live in repo-root /locales (D-CATALOG-HOME), consumed by both surfaces. `kn`
+ `hi`
34     // shipped (B5/B6, AI-drafted ? native review pending). A missing key in any locale
still falls back
35     // to `en` ? the by-design invariant.
36     const resources = {
37       en: { common: enCommon },
```

```

38     kn: { common: knCommon },
39     hi: { common: hiCommon },
40     es: { common: esCommon },
41     da: { common: daCommon },
42     id: { common: idCommon },
43   } as const;
44
45   void i18n
46     .use(ICU)
47     .use(LanguageDetector)
48     .use(initReactI18next)
49     .init({
50       resources,
51       fallbackLng: DEFAULT_LOCALE,
52       supportedLngs: [...SUPPORTED_LOCALES],
53       nonExplicitSupportedLngs: true, // en-IN -> en
54       defaultNS: "common",
55       ns: ["common"],
56       interpolation: { escapeValue: false }, // React already escapes
57       returnNull: false,
58       // Negotiation order (I18N_PLAN §2.3): persisted pref -> ?lang= -> browser -> en.
59       detection: {
60         order: ["localStorage", "querystring", "navigator"],
61         lookupLocalStorage: LANG_STORAGE_KEY,
62         lookupQuerystring: "lang",
63         caches: ["localStorage"],
64       },
65       react: { useSuspense: false },
66     });
67
68     // Keep <html lang> in sync with the active locale (screen readers + SEO use it).
Reflects the
69     // resolved content language, so an `en` fallback for an absent locale still reads as
`en`.
70     function syncDocumentLang(lng: string) {
71       if (typeof document !== "undefined") {
72         document.documentElement.lang = (i18n.resolvedLanguage || lng ||
DEFAULT_LOCALE).split("-")[0];
73       }
74     }
75     i18n.on("languageChanged", syncDocumentLang);
76     syncDocumentLang(i18n.language ?? DEFAULT_LOCALE);
77
78     export default i18n;
79

```

16. assistant (2026-06-22T09:46:59.271Z)

All confirmed against the live khelsutra tree:

- `SUPPORTED\_LOCALES` is at **line 25** (not 18; line 18 is the hiCommon import). ?
- `` sync `syncDocumentLang()` begins at **line 70**; lines 59-65 is the detection-order block. ?

Now let me look at the exact doc lines I need to edit (49, 50, 16, 68, 87, 108) precisely with current line numbers. The doc lines are stable since I just read the file. Let me apply the surgical edits.

Edit 1 ? Finding 4: `` sync citation (line 49): **i18n/index.ts:59-65** ?
i18n/index.ts:70.

17. user (2026-06-22T09:47:00.906Z)

The file C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

18. assistant (2026-06-22T09:47:07.104Z)

****Edit 2 ? Finding 3: SUPPORTED_LOCALES citation (line 50):**** ``i18n/index.ts:18`` ?
``i18n/index.ts:25``.

19. user (2026-06-22T09:47:08.074Z)

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20. assistant (2026-06-22T09:47:16.586Z)

****Edit 3 ? Finding 2: broken \$10 reference to docs/UI_PROPOSAL.md (line 108).**** The engine repo has UI docs; this khelsutra repo does not. The suggestedFix is to remove or re-point. The ``FRIENDLY_AND_MULTILINGUAL.md`` (archived) and ``engine docs/I18N_PLAN.md`` are already cited adjacently. UI_PROPOSAL.md is listed alongside ``engine docs/I18N_PLAN.md`` ? the surrounding context implies it's also an engine doc. Let me re-point it to clarify it's an engine doc rather than removing, matching the adjacent ``engine docs/I18N_PLAN.md`` framing. Let me check the exact text.

21. user (2026-06-22T09:47:17.354Z)

```
107      ## 10. References
108      **Internal `[verified]`: `web/src/i18n/{index.ts,hygiene.test.ts}`, `web/src/main.tsx`
(font-load pattern), repo-root `/locales/{en,kn,hi,es,da,id}/common.json` (hoisted by FRIENDLY
B3 ? D-CATALOG-HOME); [`FRIENDLY_AND_MULTILINGUAL.md`](archives/FRIENDLY_AND_MULTILINGUAL.md),
engine `docs/I18N_PLAN.md`, `docs/UI_PROPOSAL.md`.
109      **External `[researched]`: Unicode CLDR v48 plural-rules chart
(https://www.unicode.org/cldr/charts/48/supplemental/language\_plural\_rules.html) + plural-rules
spec; W3C BCP-47 / Chinese script subtags (`zh-Hans`/`zh-Hant`); SIL Open Font License 1.1 +
Noto CJK LICENSE (github.com/notofonts/noto-cjk); `@fontsource-variable/noto-
sans-{thai,sc,jp,kr}`; W3C Thai line-breaking (`thai-gap`); react-i18next + i18next-icu.
```

22. assistant (2026-06-22T09:47:21.065Z)

The text reads ``engine docs/I18N_PLAN.md, docs/UI_PROPOSAL.md``. The ``docs/UI_PROPOSAL.md`` is sitting in the "Internal `[verified]`" group but it's clearly meant as an engine doc (it follows ``engine docs/I18N_PLAN.md``). Let me verify it exists in the engine repo to re-point correctly rather than just remove.

23. user (2026-06-22T09:47:22.582Z)

```
docs/I18N_PLAN.md
docs/UI_PROPOSAL.md
```

24. assistant (2026-06-22T09:47:26.371Z)

``docs/UI_PROPOSAL.md`` exists in the engine repo (origin/master), same as ``docs/I18N_PLAN.md``. The audit flagged it as stale only because it's mis-grouped under "Internal `[verified]`" (which means this repo) when it's actually an engine doc ? the same situation as ``engine docs/I18N_PLAN.md`` right before it. The cleanest surgical fix that keeps the honesty tag accurate is to qualify it as an engine doc, matching the adjacent ``engine docs/I18N_PLAN.md``.

25. user (2026-06-22T09:47:28.668Z)

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26. assistant (2026-06-22T09:47:32.605Z)

****Edit 4 ? Finding 1 (medium): EX3 + "SPA-only until B3" framing (lines 87, 16, 68).**** B3 (#70) landed and site es/da/id parity (#87) shipped. The suggestedFix: mark the es/da/id portion of EX3 done (with th/zh/ja/ko remaining) and drop the "SPA-only" caveat for es/da/id in \$1/\$4/\$6.

Let me apply these. First the \$1 line (16):

27. user (2026-06-22T09:47:35.271Z)

The file C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

28. assistant (2026-06-22T09:47:37.563Z)

Now the \$6 line (68): "depends on FRIENDLY B3 landing first ? until then this is SPA-only."

29. user (2026-06-22T09:47:40.038Z)

The file C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

30. assistant (2026-06-22T09:47:45.789Z)

Now the \$4 reuse-map line. The audit's docLocation cited \$1/\$4/\$6 (lines 16, 68) and also \$4 generally. Let me check line 51 ? the catalog-home note already says B3 "hoisted them". The "SPA-only" caveat in \$4 area: the audit referenced docLocation `16,68` for the prose. \$4 line 68 is actually in \$6. Let me re-check ? line 68 is the \$6 honest-limits bullet I just edited. The \$1 is line 16. The third reference in the finding's claim was "\$1/\$4/\$6 say site localization 'depends on FRIENDLY B3 landing first'". Let me search for any remaining "SPA-only" or "depends on FRIENDLY B3" phrasing in \$4.

31. user (2026-06-22T09:47:47.110Z)

68:- `site/` marketing localization rides **FRIENDLY B3** (the shared shim) ? **B3 landed ([#70])** and **es/da/id site parity shipped ([#87])**, so the Phase-A trio is no longer SPA-only; only the **th/zh/ja/ko** site parity remains.
95:6. **Does `site/` localize for these too** for v1, or SPA-only until FRIENDLY B3 lands?
(EX3.)

32. assistant (2026-06-22T09:47:56.297Z)

Line 95 is an Open Question (\$8). The audit's finding cited \$1/\$4/\$6 (lines 16, 68) plus the EX3 tracker row (87); it did not flag \$8 Q6 specifically. However, Q6 ("SPA-only until FRIENDLY B3 lands?") is now answered/stale since B3 landed. But the audit's worklist explicitly scoped the "SPA-only" caveat to \$1/\$4/\$6 and EX3 ? it did not list line 95. To stay surgical and within the findings (no scope creep), I'll handle line 95 cautiously: it's part of the same stale "until FRIENDLY B3 lands" framing the finding targets. The finding's suggestedFix says "drop the 'until then this is SPA-only' caveat for es/da/id in \$1/\$4/\$6." \$8 Q6 is an open owner question, not a \$1/\$4/\$6 prose caveat. I'll leave Q6 as-is to avoid scope creep beyond the finding ? it's an owner-decision question, not a factual staleness claim, and the audit did not flag it.

Now the EX3 tracker row (line 87). The suggestedFix: mark es/da/id portion done with th/zh/ja/ko remaining; reflect B3 landed (#70) and #87.

33. user (2026-06-22T09:47:56.822Z)

```
85      | EX1 | **Native-speaker review** of each AI-drafted catalog (one tracked row; per-  
language sign-off) | ? | the catalog rows | **Owner (reviewer sourcing)** | ? | ? |  
86      | EX2 | **Owner-machine font eyeball** per new script (th, zh, ja, ko) ? cloud CI can't  
render fonts | ? | EB*/EC* | **Owner (owner machine)** | ? | ? |  
87      | EX3 | **`site/` marketing parity** for the 7 (rides FRIENDLY **B3** shared `t()` shim)  
| ? | FRIENDLY B3 | Blocked on B3 | ? | ? |
```

34. assistant (2026-06-22T09:48:01.617Z)

The EX3 row covers "the 7" languages. es/da/id site parity is done (#87 + B3 #70); th/zh/ja/ko remain. I'll mark it ? (In progress, the partial-done glyph in the doc's own legend), update the Gated/PR columns to reflect B3 landed and #87 shipped the Phase-A trio, with th/zh/ja/ko

remaining. The deliverable text references "the 7" ? I'll keep the row but reflect partial completion honestly.

35. user (2026-06-22T09:48:04.555Z)

The file C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md has been updated successfully. (file state is current in your context ? no need to Read it back)

36. assistant (2026-06-22T09:48:11.559Z)

Now let me also update the §9 Definition-of-done line that references EX3 with the "after FRIENDLY B3" framing ? actually, line 104 says "(if in scope)* `site/` marketing renders the 7 via the shared shim (EX3, after FRIENDLY B3)." That's a DoD checkbox, not flagged in the findings. The finding scoped to EX3 (tracker row 87) + §1/§4/§6 prose. Line 104 references "after FRIENDLY B3" which is now landed, but it's a DoD criterion (the full 7 still aren't done on the site), so the box remaining unchecked is still honest. I'll leave it ? the "after FRIENDLY B3" is a dependency note that's harmlessly satisfied; editing it would be scope creep beyond the findings.

Let me verify my EX3 edit and the changelog don't need a status-header bump. The audit's perDoc summary said the doc's header (IN PROGRESS, Phase A shipped, B/C owner-gated) is **correct** ? no header change needed. Let me confirm I haven't missed anything by reviewing the final state of the edited regions.

37. user (2026-06-22T09:48:12.442Z)

```
48      **Reuse map (what's reused vs new):**
49      - *Reused unchanged:* the negotiation contract (persisted pref ? `?lang=` ? navigator ?
`en`), the ICU catalog format, the `@fontsource` `unicode-range`-scoped self-host pattern
(`main.tsx:7-8`), the key-parity + no-bare-strings + offline CI (`hygiene.test.ts`), the
draft?review pipeline, the `<html lang>` sync (`i18n/index.ts:70` `syncDocumentLang`).
50      - *New:* 7 `common.json` catalogs; `SUPPORTED_LOCALES` entries (`i18n/index.ts:25`); **4
new Noto font imports**; an es 3-form plural allowance in the catalog + tooling; a CJK
lazy-load + offline-subset strategy (Phase C); Thai line-break CSS safety net.
51      - ? Catalog-home (resolved by FRIENDLY B3): catalogs now live at repo-root
**/locales/{en,kn,hi,es,da,id}/common.json ? the shared top-level dir from the I18N_PLAN
"D-CATALOG-HOME" diagram. FRIENDLY B3 hoisted them out of `web/src/i18n/locales/` (the SPA
re-points its imports there; the `site/` shim consumes a `copy:locales` mirror). These es/da/id
catalogs moved with the hoist, as anticipated.
```

38. assistant (2026-06-22T09:48:17.711Z)

All four stale findings are now applied. Let me do a final review of the complete set of edits to confirm correctness and that I haven't introduced any markdown issues.

39. user (2026-06-22T09:48:18.350Z)

```
16      Concrete outcome ("good"): a user whose browser is in any of the 7 languages lands
on the Studio (and the marketing pages too ? FRIENDLY B3 landed [#70], with es/da/id site parity
shipped [#87]; th/zh/ja/ko site parity still pending), and ? without help ? understands the
product, runs the tool, and gets a reel, entirely in their language, with the fonts
rendering crisply offline.
```

40. assistant (2026-06-22T09:48:21.431Z)

All edits are in place and verified against the live repos. Here is my final report.

41. user (2026-06-22T09:48:41.653Z)

Structured output provided successfully

42. assistant (2026-06-22T09:48:48.402Z)

All four stale findings for `docs/MULTILINGUAL\_EXPANSION.md` have been re-verified against the live khelsutra and engine repos and fixed surgically. Summary:

- ****EX3 / "SPA-only until B3"**** (medium) ? B3 (#70) and es/da/id site parity (#87) shipped; updated §1, §6, and the §7 EX3 tracker row (???) to reflect that the Phase-A trio is no longer SPA-only, with th/zh/ja/ko remaining.
- ****Broken §10 reference**** (low) ? `docs/UI\_PROPOSAL.md` doesn't exist in khelsutra but does on engine origin/master; regrouped it under the engine-docs citation.
- ****Two line-number drifts**** (low) ? `SUPPORTED\_LOCALES` :18?:25 and `` sync :59-65?:70, both confirmed against `web/src/i18n/index.ts`.

Two non-flagged items were skipped to avoid scope creep (§8 Q6's "until B3 lands" open-question framing, and the dated "168 web tests green" changelog note the audit marked correct/optional). No code changes were required.